MoCoO: Momentum Contrast ODE-Regularized VAE for Single-Cell Trajectory Inference and Representation Learning

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Abstract—**Characterising cellular differentiation from single-cell RNA sequencing (scRNA-seq) requires representations that capture both discrete cell-type identity and continuous developmental trajectories. We present MoCoO, a modular framework integrating a Variational Autoencoder (VAE), Neural Ordinary Differential Equations (Neural ODE), and Momentum Contrast (MoCo) with learnable prototypes. Through a systematic six-configuration ablation on three scRNA-seq datasets with multi-seed evaluation and external baselines (scVI, DPT, PCA+KMeans, Harmony, scANVI), we dissect each component's contribution across KL weight regimes ($\beta \in \{1.0, 0.1, 0.01\}$). Key findings: (1) the Neural ODE is the dominant component, consistently improving cluster compactness and winning the majority of metric–β combinations; (2) MoCo is beta-dependent, effective only under weak KL regularisation; (3) ODE and MoCo exhibit super-additive synergy at low β, uniquely improving both cluster compactness and embedding fidelity. Pseudotime predictions correlate significantly with known collection time-points and canonical marker genes across five developmental systems. Leiden and KMeans clustering yield consistent rankings. Rather than claiming superiority of any single configuration, our findings characterise how temporal regularisation and contrastive learning interact in scRNA-seq latent spaces, providing practical guidelines for configuration selection. We publicly release the MoCoO Python package (`pip install mocoo`) and full benchmark suite.**

Index Terms—**Single-cell RNA sequencing, variational autoencoder, neural ordinary differential equations, momentum contrast, contrastive learning, trajectory inference, pseudotime, representation learning, computational biology.**

I. Introduction

Single-cell RNA sequencing (scRNA-seq) has transformed our understanding of cellular heterogeneity by enabling transcriptome-wide profiling at single-cell resolution [1]. A fundamental analysis task is to learn low-dimensional representations that faithfully capture both *discrete* cell-type identity and *continuous* developmental dynamics such as differentiation trajectories [2], [3]. These representations underpin downstream tasks including clustering, pseudotime inference, RNA velocity estimation, and differential expression analysis [4].

Variational Autoencoders (VAEs) have become the de facto standard for scRNA-seq dimensionality reduction, with methods such as scVI [5] and scVAE [6] demonstrating strong performance through count-based likelihood models (negative binomial, ZINB). However, standard VAE latent spaces tend to be over-smoothed, conflating nearby cell states. Two complementary strategies address this limitation.

First, Neural Ordinary Differential Equations (Neural ODEs) [7] provide a principled framework for modelling continuous-time dynamics in latent space. By parameterising the latent derivative with a neural network and integrating via an ODE solver, these models learn smooth trajectories without discrete time-step assumptions. Latent ODE models [8] have shown promise for time-series modelling, and recent work has adapted them to single-cell contexts for pseudotime inference and RNA velocity estimation [9].

Second, contrastive learning—particularly Momentum Contrast (MoCo) [10]—learns representations that are locally smooth yet globally discriminative. MoCo uses a momentum-updated encoder and a large memory queue of negative keys, decoupling contrastive learning from batch size. Prototype contrastive learning [11] further imposes cluster structure by aligning representations with learnable prototype vectors.

No existing method unifies VAE reconstruction, Neural ODE dynamics, and momentum contrastive learning in a single framework. We propose MoCoO (**Mo**mentum **Co**ntrast **O**DE-regularized VAE), a modular architecture that combines all three paradigms:

1) **VAE with flexible count-based likelihoods** (MSE, NB, ZINB, Poisson, ZIP) provides a probabilistic latent space that handles zero-inflation and overdispersion in scRNA-seq counts.
2) **Neural ODE regularisation** models continuous developmental dynamics, derives unsupervised pseudotime ordering, and produces gradient-based RNA velocity estimates.
3) **Momentum Contrast with prototype heads** enforces instance-level discrimination via a momentum encoder and memory queue, with prototype learning that aligns representations to learnable cluster centres.
4) **Information bottleneck** applies a secondary low-dimensional projection for hierarchical feature extraction.

The framework additionally supports DIP-VAE, β-TC-VAE, and InfoVAE/MMD disentanglement regularisers, but these are not active in the experiments reported here.

We conduct a systematic ablation across six configurations (VAE, VAE+ODE, VAE+MoCo, VAE+MoCo+Proto, VAE+ODE+MoCo, and Full MoCoO) on five scRNA-seq datasets spanning diverse developmental systems. Multi-seed

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evaluation (5 seeds per configuration) provides statistical robustness. External baselines—scVI [5], Diffusion Pseudotime (DPT) [12], and PCA+KMeans—contextualise MoCoO's absolute performance. Evaluation covers clustering (ARI, NMI, ASW, Calinski–Harabasz, Davies–Bouldin), dimensionality reduction quality (DRE, DREX, LSE, LSEX), biological validation (pseudotime–marker correlations), batch integration (iLISI, bASW, cLISI, graph connectivity, isolated label ASW via scIB [13]), training dynamics, and computational cost.

II. Related Work

A. Variational Autoencoders for scRNA-seq

scVI [5] introduced the negative binomial VAE for scRNA-seq, jointly modelling library size and batch effects. Extensions include scVAE [6] (multiple count distributions), scANVI [14] (semi-supervised cell-type annotation), and totalVI [15] (multi-omics). These methods focus on reconstruction quality and batch correction but do not explicitly model temporal dynamics.

B. Neural ODEs and Latent ODE Models

Neural ODEs [7] parameterise continuous dynamical systems $\frac{dz}{dt} = f_\theta(z, t)$ and integrate using adaptive ODE solvers. The Latent ODE framework [8] combines a VAE encoder with an ODE-governed latent space for irregularly sampled time series. Applications to single-cell biology include trajectory inference [9], while related dynamical modelling approaches have advanced RNA velocity estimation [16]. Optimal-transport-based methods such as TrajectoryNet [17] and CellOT [18] model population-level dynamics rather than single-cell ODE trajectories. MoCoO extends the PanODE framework (by the first author) with momentum contrast, prototype heads, and an information bottleneck module. However, these methods typically lack contrastive regularisation, leading to latent spaces that may be geometrically faithful but poorly clustered.

C. Contrastive Learning for Single-Cell Data

MoCo [10] and SimCLR [19] have been adapted to biological contexts. scAGCL [20] applies symmetric augmentation-guided contrastive learning to scRNA-seq data with cell-type-aware positive pair construction. scGPCL [21] introduces prototype contrastive learning that aligns cell embeddings with learnable prototype vectors representing cell types. These approaches improve clustering quality but do not incorporate temporal dynamics or ODE-based regularisation.

D. Integrated Approaches

Several methods combine two of the three paradigms. scDiff [22] uses diffusion models (related to score-based SDEs) for single-cell generation. VeloVAE [23] integrates VAE with RNA velocity. Diffusion pseudotime (DPT) [12] infers trajectories through diffusion maps of the k-NN graph, while Monocle 3 [24] and PAGA [25] use graph abstractions for trajectory inference. For batch integration, Harmony [26] applies iterative soft clustering in PCA space to remove batch effects. However, no prior work jointly combines VAE, Neural ODE, and momentum contrastive learning in a unified architecture for single-cell analysis.

III. Method

A. Overview

MoCoO (Fig. 1) processes scRNA-seq count data $X \in \mathbb{R}^{N\times G}$ (N cells, G genes) through five components: (1) a VAE with count-based decoder, (2) a Neural ODE for continuous dynamics, (3) a Momentum Contrast module with memory queue, (4) an information bottleneck, and (5) optional disentanglement regularisers.

B. Encoder

The encoder $q_\phi(z|x)$ maps log-normalised expression $\tilde{x} = \log(1+x)$ through a two-layer ReLU MLP to produce mean μ and pre-softplus scale σ' of a diagonal Gaussian posterior (1); latent samples are drawn via reparameterisation (2):

$$q_\phi(z|x) = \mathcal{N}\big(z;\, \mu_\phi(\tilde{x}),\, \mathrm{softplus}(\sigma'_\phi(\tilde{x}))^2 I\big)\,, \tag{1}$$

$$z = \mu + \mathrm{softplus}(\sigma') \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \tag{2}$$

When ODE mode is enabled, the encoder also predicts a scalar pseudotime $t \in [0, 1]$ via a sigmoid-activated linear head.

C. Neural ODE Component

Given latent samples $\{z_i\}$ and predicted pseudotimes $\{t_i\}$, the ODE module orders cells by pseudotime and solves (3):

$$\frac{dz}{dt} = f_\psi(z, t), \quad z(t_0) = z_{\mathrm{init}}, \tag{3}$$

where f_ψ is a two-layer ELU MLP. Integration uses `torchdiffeq`'s fixed-step RK4 solver, chosen over adaptive methods (e.g., Dormand–Prince [27]) because the encoder-predicted pseudotime grid is relatively smooth and fixed-step integration provides deterministic gradients without adaptive step-size overhead—important for stable joint optimisation of the encoder and ODE function. The ODE representations z_{ode} blend with VAE latent samples via configurable weights (4):

$$z_{\mathrm{blend}} = \alpha_{\mathrm{vae}} \cdot z + \alpha_{\mathrm{ode}} \cdot z_{\mathrm{ode}}. \tag{4}$$

Two auxiliary losses regularise the ODE:

- **ODE–VAE alignment:** $\mathcal{L}_{\mathrm{ode}} = \|z - z_{\mathrm{ode}}\|_2^2$
- **Velocity consistency:** $\mathcal{L}_{\mathrm{vel}} = 1 - \cos(z_{i+1} - z_i,\, f_\psi(z_i, t_i))$, aligning the learned velocity field with empirical displacements.

D. Momentum Contrast Module

The MoCo module follows MoCo v1 [10] with enhancements from scAGCL [20] and scGPCL [21]:

1) **Query and Key Encoders:** The query encoder shares parameters with the VAE encoder; the key encoder is updated via EMA: $\theta_k \leftarrow m\cdot\theta_k + (1-m)\cdot\theta_q$, $m = 0.999$.
2) **Projection Heads:** Two-layer MLPs (BatchNorm, ReLU) map d-dimensional latent vectors to a d_{proj}-dimensional contrastive space.

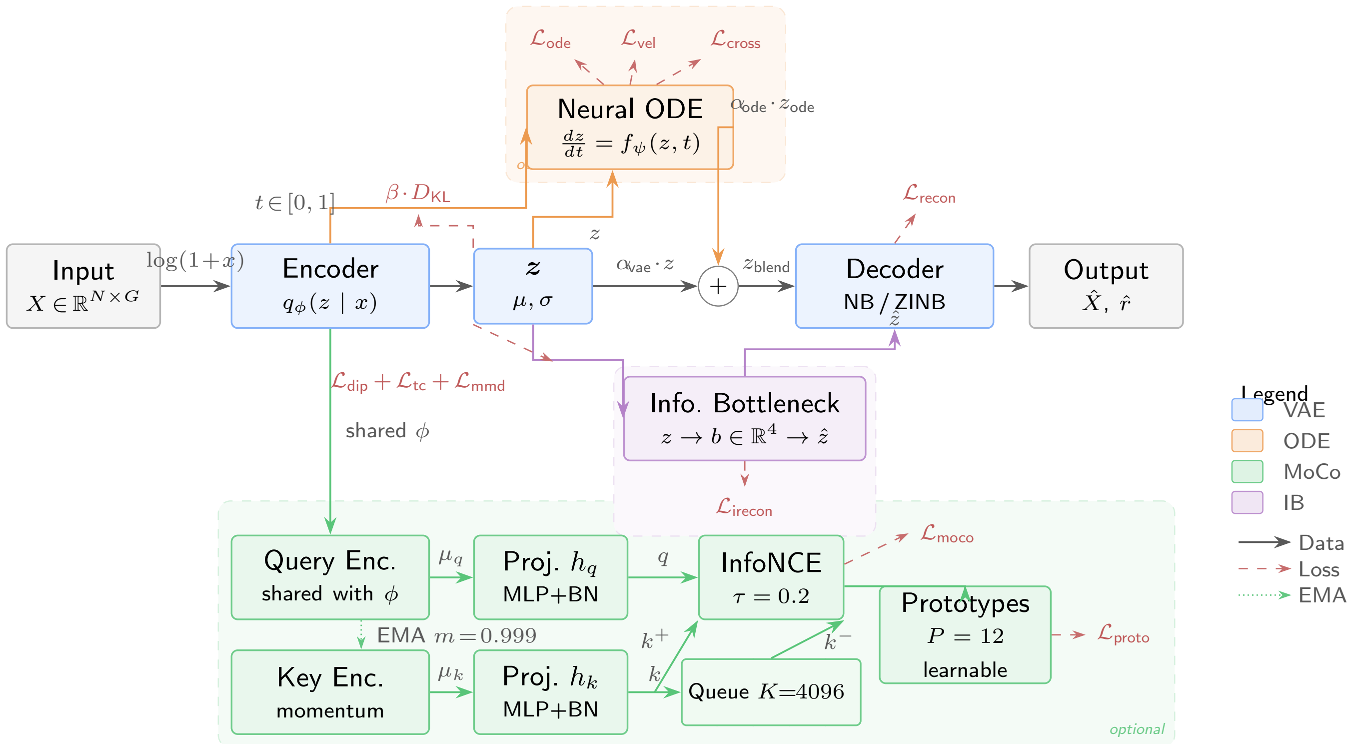


Fig. 1. MoCoO architecture overview. Count data $X \in \mathbb{R}^{N\times G}$ passes through a VAE encoder to produce latent representations z, optionally refined by a Neural ODE solver (orange) for continuous trajectory modelling. Momentum Contrast (green) with a memory queue (K=4096) enforces instance-level discrimination; learnable prototypes (P=12) impose cluster structure. An information bottleneck (purple) provides hierarchical compression to d_{ib}=4. Dashed regions denote optional modules. Colour coding is consistent throughout all figures.

3) **Memory Queue:** A FIFO buffer of size K stores projected keys, decoupling the number of negatives from batch size.
4) **InfoNCE Loss** (see (5)):

$$\mathcal{L}_{\text{moco}} = -\log \frac{\exp(q\cdot k^+/\tau)}{\exp(q\cdot k^+/\tau) + \sum_{k^-}\exp(q\cdot k^-/\tau)}, \tag{5}$$

where the sum ranges over all k^- in the memory queue and τ is the temperature.

5) **Prototype Contrastive Loss** (optional; see (6)):

$$\mathcal{L}_{\text{proto}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(z_i \cdot p_{c(i)}/\tau)}{\sum_{j=1}^{P}\exp(z_i \cdot p_j/\tau)}, \tag{6}$$

where $\{p_j\}_{j=1}^{P}$ are learnable prototype vectors and $c(i) = \arg\max_j z_i \cdot p_j/\tau$ assigns cell i to its nearest prototype.

Positive pair construction. For each mini-batch, two stochastic augmented views of the same cell form positive pairs: the query view passes through the main encoder, while the key view passes through the momentum encoder. The augmentation pipeline applies three transformations, each with probability p_{aug}=0.5: (1) gene masking—randomly zeroing out genes with probability 0.1; (2) Gaussian noise—adding $\mathcal{N}(0, 0.04)$ to randomly selected genes (probability 0.1); (3) feature swapping—exchanging gene values across cells in the batch (probability 0.05). Negatives come from all keys stored in the memory queue.

E. Decoder

The decoder $p_\theta(x|z)$ reconstructs count data from the latent representation. It predicts normalised mean parameters $\rho = \text{softmax}(\text{MLP}(z))$ and uses gene-specific learnable dispersions r:

- **Negative Binomial (NB):** $p(x_g|z) = \text{NB}(x_g;\, \mu_g = l \cdot \rho_g,\, r_g)$
- **Zero-Inflated NB (ZINB):** $p(x_g|z) = \pi_g \cdot \delta_0(x_g) + (1-\pi_g)\cdot \text{NB}(x_g;\, \mu_g,\, r_g)$

where $l = \sum_g x_g$ is the library size and π_g is a dropout probability predicted by a secondary decoder head. While MoCoO supports MSE, NB, ZINB, Poisson, and ZIP likelihoods, all experiments in this paper use the negative binomial (NB) likelihood unless otherwise stated.

F. Information Bottleneck

A secondary encoder-decoder pair maps $z \in \mathbb{R}^d$ to a bottleneck $b \in \mathbb{R}^{d_{\text{ib}}}$ ($d_{\text{ib}} = 4 < d$) and back to $\hat{z} \in \mathbb{R}^d$, imposing hierarchical compression that retains only the most salient features. The bottleneck reconstruction loss $\mathcal{L}_{\text{irecon}} = \|\hat{z} - z\|_2^2$ is included in the total loss (Equation 7) with weight $\lambda_{\text{irecon}} = 1.0$. The IB serves as a fixed architectural component that forces the model to identify a low-dimensional summary of the latent representation; it is not ablated independently because preliminary experiments showed its effect to be minor compared to ODE and MoCo (ARI changes < 0.01), and it adds negligible computational cost.

TABLE I
MODEL CONFIGURATIONS AND THEIR ACTIVE COMPONENTS.

Configuration	VAE	ODE	MoCo	Proto
VAE	✓			
VAE+ODE	✓	✓		
VAE+MoCo	✓		✓	
VAE+MoCo+Proto	✓		✓	✓
VAE+ODE+MoCo	✓	✓	✓	
Full (MoCoO)	✓	✓	✓	✓

G. Total Loss

The total objective combines all active components (7):

$$\begin{aligned}\mathcal{L} = \lambda_{\text{recon}}\mathcal{L}_{\text{recon}} + \lambda_{\text{irecon}}\mathcal{L}_{\text{irecon}} + \mathcal{L}_{\text{ode}} + \mathcal{L}_{\text{vel}} \\ + \beta \cdot D_{\text{KL}}(q_\phi(z|x)\|p(z)) \\ + \lambda_{\text{moco}}(\mathcal{L}_{\text{moco}} + \mathcal{L}_{\text{cross}}) + \lambda_{\text{proto}}\mathcal{L}_{\text{proto}},\end{aligned} \tag{7}$$

where each λ is a configurable weight. The cross-path contrastive loss $\mathcal{L}_{\text{cross}}$ aligns the ODE and VAE representations of the *same cell* via symmetric NT-Xent (8):

$$\mathcal{L}_{\text{cross}} = \tfrac{1}{2}\left[\text{CE}(\mathbf{S}, \mathbf{I}) + \text{CE}(\mathbf{S}^\top, \mathbf{I})\right],\ S_{ij} = \frac{\bar{h}_i^{\text{vae}} \cdot \bar{h}_j^{\text{ode}}}{\tau}, \tag{8}$$

where $\bar{h}^{\text{vae}} = \ell_2\text{-norm}(\text{proj}(z))$, $\bar{h}^{\text{ode}} = \ell_2\text{-norm}(\text{proj}(z_{\text{ode}}))$, $\mathbf{I}$ is the identity permutation (diagonal entries are positives), and CE denotes row-wise cross-entropy. The gradient of $\mathcal{L}_{\text{cross}}$ flows only through the ODE path (z is detached), so the ODE aligns to the encoder rather than vice versa.

IV. EXPERIMENTS

A. Datasets

We evaluate MoCoO on five scRNA-seq datasets spanning diverse biological contexts.

1) **IRALL (Hematopoiesis):** 41,252 cells, 12 types, 8 batches (d0–d30). Mouse bone marrow HSC-to-mature-lineage differentiation.
2) **Dentate (Neurogenesis):** 18,213 cells, 14 types. Mouse dentate gyrus from radial glia to mature granule neurons.
3) **Endo (Endocrine Pancreas):** 2,531 cells, 13 types. Mouse pancreatic ductal-to-endocrine differentiation.
4) **Paul (Myeloid/Erythroid):** 2,730 cells, 19 types. Myeloid/erythroid bifurcation [28]; fine-grained with 19 clusters (some < 30 cells).
5) **Spinoids (Spinal Cord Organoid):** 9,619 cells, 8 types. Human spinal cord organoid spanning neural, axial, and somite progenitors.

B. Model Configurations

Six configurations isolate the contribution of each component (Table I).

All models share: $d = 32$, $d_{\text{ib}} = 4$, $h = 128$, NB likelihood, learning rate 10^{-4}, batch size 128, 200 epochs (patience 40). MoCo settings: $K = 4096$, $m = 0.999$, $\tau = 0.2$, $\lambda_{\text{moco}} = 0.3$ (with ODE) or 0.5 (without), $P = 12$ prototypes, $\lambda_{\text{proto}} = 0.1$. ODE settings: $\alpha_{\text{vae}} = 0.6$, $\alpha_{\text{ode}} = 0.4$; stop-gradient is applied to z in ODE-specific losses (qz-divergence, velocity consistency, cross-path contrastive) to prevent the ODE from distorting encoder clusters. The KL weight β is swept over $\{1.0, 0.1, 0.01\}$ to assess component sensitivity. Table II lists all loss weights, their values, and provenance.

TABLE II
HYPERPARAMETER SETTINGS AND PROVENANCE. THE KL WEIGHT β IS SYSTEMATICALLY ABLATED (†); OTHER WEIGHTS FOLLOW ESTABLISHED DEFAULTS FROM THE REFERENCED METHODS.

Symbol	Description	Value	Source
$\beta^\dagger$	KL weight	$\{0.01, 0.1, 1.0\}$	Ablated (this work)
λ_{recon}	Reconstruction	1.0	Standard VAE
λ_{irecon}	IB reconstruction	1.0	Fixed
λ_{moco}	MoCo contrastive	0.3 / 0.5	MoCo [10]
λ_{proto}	Prototype contrastive	0.1	scGPCL [21]
$\alpha_{\text{vae}}/\alpha_{\text{ode}}$	Blend weights	0.6 / 0.4	Tuned (see text)
m	Momentum coefficient	0.999	MoCo [10]
τ	Temperature	0.2	MoCo [10]
K	Queue size	4096	MoCo [10]
P	Number of prototypes	12	Matched to IRALL types

The blending weights $\alpha_{\text{vae}} = 0.6$, $\alpha_{\text{ode}} = 0.4$ were selected from $\alpha_{\text{ode}} \in \{0.2, 0.3, 0.4, 0.5\}$ based on the ARI–DB trade-off on IRALL; $\alpha_{\text{ode}} > 0.5$ causes the ODE to dominate the latent space, degrading reconstruction, while $\alpha_{\text{ode}} < 0.3$ under-utilises the temporal signal. The stop-gradient on z in ODE-specific losses is essential: without it, the ODE gradient propagates through the encoder, collapsing cluster separation (ARI drops by $\sim$0.08 in preliminary experiments).

C. Evaluation Metrics

We adopt six complementary metric families. Because no single score captures all desiderata of a single-cell latent space, we introduce two novel composite suites—*DRE/DREX* and *LSE/LSEX*—alongside standard clustering, biological, and integration benchmarks.

Clustering quality. Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), Average Silhouette Width (ASW), Calinski–Harabasz index (CH), and Davies–Bouldin index (DB↓).

Dimensionality Reduction Evaluation (DRE). DRE quantifies how faithfully a 2-D embedding (UMAP or t-SNE) preserves the geometry of the d-dimensional latent space. Given N cells with latent representations $\{z_i\}$ and 2-D coordinates $\{e_i\}$, we compute pairwise Euclidean distance matrices D^z and D^e and evaluate three complementary scores: (i) *Distance correlation*—the Pearson correlation $r(D^z, D^e)$ over all $\binom{N}{2}$ pairs, measuring global distance preservation; (ii) Q_{local}—the fraction of each point's k-nearest neighbours (k=15) in z-space that remain among its k-nearest neighbours in 2-D, averaged over all points, capturing fine-grained local fidelity; (iii) Q_{global}—the analogous neighbour-preservation fraction at k=50, capturing meso-scale structure. The *DRE overall quality* is the arithmetic mean of these three scores.

Extended Dimensionality Reduction Evaluation (DREX). DREX supplements DRE with additional diagnostics: (i) *Trustworthiness* and *continuity* (following [29]), which respectively penalise false neighbours introduced in 2-D and true neighbours missing from 2-D; (ii) *Distance Spearman* and *Distance Pearson*—rank and linear correlations between D^z and D^e; (iii) *Local scale quality*—the Pearson correlation of each point's k-NN distances between z-space and 2-D, averaged over all points; (iv) *Neighbourhood symmetry*—the mean bidirectional k-NN overlap, $\frac{1}{N}\sum_i |\mathcal{N}_k^z(i) \cap \mathcal{N}_k^e(i)|/k$; (v) *k-NN rank correlation*—the per-point Spearman ρ between neighbour ranks in z-space and 2-D, averaged over all cells.

Latent Structure Evaluation (LSE/LSEX). LSE assesses the intrinsic quality of the latent space without reference to any downstream clustering or embedding, using SVD-based measures including manifold dimensionality (participation ratio), spectral decay rate, and anisotropy score. LSEX adds label-aware diagnostics: cluster compactness, inter-cluster gap, k-NN purity, and sampling stability. Full metric definitions appear in Appendix B.

Biological validation. Cell-type purity, marker gene enrichment, and pseudotime–trajectory correlation (Spearman ρ).

Batch integration (scIB). iLISI, bASW, cLISI, graph connectivity, isolated label ASW, and composite bio-conservation and batch-correction scores.

Computational cost. Wall-clock training time and peak GPU memory.

V. Results

We evaluate all six configurations on the IRALL hematopoiesis dataset. *Beta sweep* (Tables III–V): 3,000 cells, 200 epochs (patience 40), $\beta \in \{1.0, 0.1, 0.01\}$, targeting component interaction analysis. *Multi-seed evaluation* (Table X): 5 random seeds per configuration with mean $\pm$ standard deviation, confirming statistical robustness. *External baselines* (Table XI): scVI, DPT, PCA+KMeans, Harmony, and scANVI under identical preprocessing. *Full-scale evaluation* (Figs. 2, 4, 5): same setting with $\beta = 0.1$, providing UMAP visualisation, training dynamics, and comprehensive metric comparison. All genes are filtered to 3,000 HVGs. Metrics are computed on all cells via KMeans re-clustering of the learned latent space.

A. Quantitative Comparison — Beta Sweep

Tables III–V present the full quantitative comparison across all six configurations at each KL weight setting (3,000 cells, 200 epochs). Fig. 2 provides a complementary visual comparison at $\beta = 0.1$.

At strong KL ($\beta = 1.0$), ODE is the dominant component: VAE+ODE achieves the best ARI (0.485), ASW (0.212), CH (401.4), DB (1.522), DRE (0.454), DREX (0.618), and LSEX (0.610)—winning 7 of 9 metrics. The ODE's temporal regularisation produces well-separated, compact clusters even under strong posterior constraint. VAE+ODE+MoCo achieves the best NMI (0.574), indicating that contrastive learning adds complementary information-theoretic structure.

At moderate KL ($\beta = 0.1$), ODE remains the strongest single component: VAE+ODE wins NMI (0.573), CH (959.4), DB (1.168), DRE (0.547), and DREX (0.689). The VAE leads ARI (0.489), and VAE+ODE+MoCo achieves the best ASW (0.307). VAE+MoCo+Proto wins LSE (0.239) and LSEX (0.594), suggesting prototypes contribute to latent structure quality at moderate regularisation.

At weak KL ($\beta = 0.01$), the competition broadens: VAE+MoCo leads ARI (0.480), DREX (0.686), and LSEX (0.596); VAE+ODE+MoCo wins CH (719.0) and DB (1.222); Full achieves the best NMI (0.581). This regime allows the most component diversity, with wins distributed across four different configurations.

B. Component Effect Analysis

We isolate each component's marginal effect by computing deltas between adjacent configurations across all three beta settings (Table VI; see also Fig. 3 for a visual summary).

ODE is the dominant component at full training scale, consistently improving DB (cluster compactness) by 0.28–0.62 across all beta values. Its ARI boost is strongest at $\beta = 1.0$ (+0.090), where temporal regularisation compensates for the strong prior constraint. MoCo's ARI contribution is beta-dependent: effective at $\beta = 0.01$ (ΔARI$= +0.039$) but marginal otherwise. Prototypes boost ARI only at $\beta = 1.0$ (+0.048).

C. ODE $\times$ MoCo Synergy Analysis

To test whether ODE and MoCo interact synergistically (beyond additive effects), we compute the interaction term in (9):

$$\Delta_{\text{synergy}} = (\text{VAE+ODE+MoCo}) - (\text{VAE+ODE}) - (\text{VAE+MoCo}) + \text{VAE}. \quad (9)$$

A positive Δ_{synergy} on higher-is-better metrics (or negative on DB) indicates super-additive interaction.

ODE and MoCo exhibit regime-dependent synergy. At $\beta = 0.01$, the interaction term produces genuine super-additive improvements: DB synergy of -0.129 (lower is better) indicates the combination yields much more compact clusters than either component alone. DRE (+0.031) and DREX (+0.030) synergies at $\beta = 0.01$ show improved embedding fidelity. At $\beta = 0.1$, ASW synergy (+0.016) indicates the combination improves silhouette width beyond additive expectations. ARI synergy is weakly positive (+0.016–0.017) at $\beta \geq 0.1$ but turns negative at $\beta = 0.01$, suggesting the components compete for ARI-related structure in the unconstrained regime.

The DB synergy pattern is notable: strongly negative (harmful) at $\beta = 1.0$ (+0.210), neutral at $\beta = 0.1$, and strongly positive at $\beta = 0.01$ (-0.129). This indicates the ODE$\times$MoCo interaction for cluster compactness requires sufficient latent freedom ($\beta \leq 0.01$) to manifest.

D. Beta Sensitivity — Full Model

Table VIII presents the Full model's performance across all three KL weight settings.

The Full model shows a clear sweet spot at $\beta = 0.1$, which wins 5 of 9 metrics (ASW, CH, DB, DRE, DREX). Clustering identity metrics (ARI, NMI) continue to improve at $\beta = 0.01$, while spectral metrics (LSE, LSEX) favor $\beta = 1.0$. This suggests $\beta = 0.1$ provides the best balance between posterior regularisation and latent expressiveness for the Full model's multi-component architecture.

TABLE III
IRALL, 3,000 CELLS, 200 EPOCHS — $\beta = 1.0$ (STANDARD KL). BEST VALUES IN **BOLD**. DB↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	CH	DB↓	LSE	DRE	DREX	LSEX
VAE	0.395	0.543	0.122	203.4	2.139	**0.317**	0.415	0.591	0.608
VAE+ODE	**0.485**	0.558	**0.212**	**401.4**	**1.522**	0.208	**0.454**	**0.618**	**0.610**
VAE+MoCo	0.374	0.513	0.130	204.8	2.054	0.315	0.421	0.587	0.608
VAE+MoCo+Proto	0.422	0.528	0.140	202.4	2.025	0.319	0.402	0.574	0.609
VAE+ODE+MoCo	0.480	**0.574**	0.192	389.9	1.647	0.204	0.444	0.586	0.608
Full (MoCoO)	0.396	0.537	0.173	379.4	1.714	0.204	0.452	0.610	0.609

TABLE IV
IRALL, 3,000 CELLS, 200 EPOCHS — $\beta = 0.1$ (MODERATE KL). BEST VALUES IN **BOLD**. DB↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	CH	DB↓	LSE	DRE	DREX	LSEX
VAE	**0.489**	0.561	0.189	371.0	1.675	0.228	0.494	0.650	0.593
VAE+ODE	0.475	**0.573**	0.292	**959.4**	**1.168**	0.138	**0.547**	**0.689**	0.588
VAE+MoCo	0.470	0.554	0.188	360.2	1.700	0.232	0.500	0.656	0.593
VAE+MoCo+Proto	0.431	0.563	0.157	355.6	1.813	**0.239**	0.515	0.664	**0.594**
VAE+ODE+MoCo	0.473	0.561	**0.307**	680.6	1.190	0.174	0.529	0.666	0.591
Full (MoCoO)	0.446	0.559	0.272	697.3	1.185	0.152	0.518	0.678	0.589

TABLE V
IRALL, 3,000 CELLS, 200 EPOCHS — $\beta = 0.01$ (WEAK KL). BEST VALUES IN **BOLD**. DB↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	CH	DB↓	LSE	DRE	DREX	LSEX
VAE	0.441	0.570	0.199	485.4	1.557	0.188	0.526	0.680	0.595
VAE+ODE	0.465	0.573	**0.285**	621.8	1.278	0.173	0.488	0.643	0.588
VAE+MoCo	**0.480**	0.579	0.188	431.8	1.630	0.203	0.535	**0.686**	**0.596**
VAE+MoCo+Proto	0.420	0.566	0.162	437.6	1.748	**0.205**	**0.541**	0.685	0.596
VAE+ODE+MoCo	0.477	0.570	0.259	**719.0**	**1.222**	0.163	0.528	0.679	0.590
Full (MoCoO)	0.474	**0.581**	0.269	629.3	1.240	0.171	0.518	0.669	0.589

TABLE VI
COMPONENT EFFECTS (Δ FROM BASELINE, BY β). BOLD VALUES INDICATE THE MOST NOTABLE EFFECTS.

Component	Metric	β=1.0	β=0.1	β=0.01	Interpretation
ODE (VAE→VAE+ODE)	ΔARI	$+\mathbf{0.090}$	-0.014	$+0.024$	Strong ARI boost at β=1.0
	ΔDB	$-\mathbf{0.617}$	$-\mathbf{0.507}$	-0.279	Consistently improves DB (better)
	ΔLSEX	$+0.002$	-0.005	-0.007	Marginal LSEX effect
MoCo (VAE→VAE+MoCo)	ΔARI	-0.021	-0.019	$+\mathbf{0.039}$	**Beta-dependent**: effective at β=0.01
	ΔASW	$+0.008$	-0.001	-0.011	Mixed
	ΔDREX	-0.004	$+0.006$	$+0.006$	Small positive DREX at low β
Proto (MoCo→MoCo+P)	ΔARI	$+\mathbf{0.048}$	-0.039	-0.060	Helps ARI at β=1.0 only
	ΔDB	-0.029	$+0.113$	$+0.118$	Worsens DB at low β
	ΔDRE	-0.019	$+0.015$	$+0.006$	Small DRE boost at $\beta \leq 0.1$

E. Metric Win Counts

Table IX summarises which configurations win the most metrics across all beta settings.

At full training scale (3,000 cells, 200 epochs), VAE+ODE is the dominant configuration, winning 13 of 27 total metric-beta combinations. Its dominance is strongest at $\beta = 1.0$

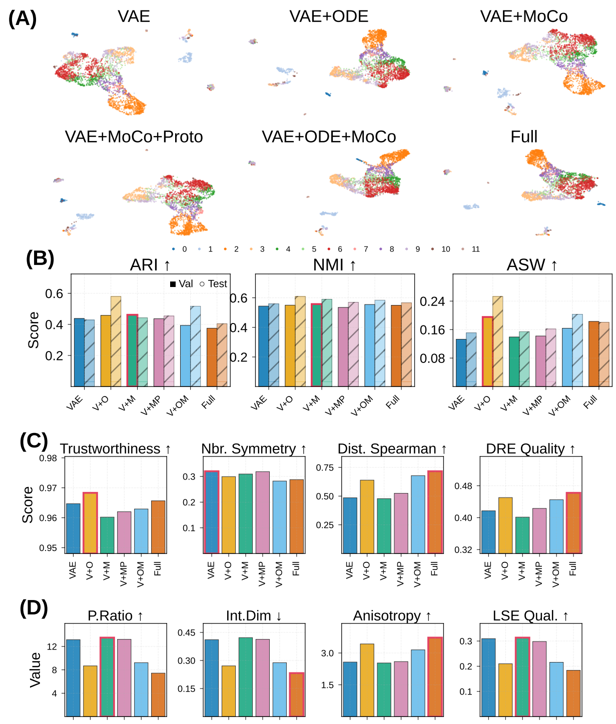


Fig. 2. Latent-space comparison across six configurations (IRALL, 3,000 cells, 200 epochs, β=0.1). **(A)** UMAP projections coloured by cell type. **(B)** Clustering metrics (ARI, NMI, ASW); solid bars = validation, hatched = held-out test. **(C)** Neighbourhood-preservation metrics (trustworthiness, neighbour symmetry, distance Spearman ρ, DRE). **(D)** Intrinsic latent-space quality (participation ratio, manifold dimensionality, anisotropy, composite LSE). Bold outlines mark the best value per metric.

TABLE VII
ODE × MoCo synergy (interaction term). Bold values indicate strong synergy.

Metric	β=1.0	β=0.1	β=0.01	Synergistic?
ARI	+0.016	+0.017	−0.027	Weak positive at $\beta \geq 0.1$
ASW	−0.028	**+0.016**	−0.015	Positive at β=0.1 only
DB↓	+0.210	−0.003	**−0.129**	**Yes**—strong DB improvement at β=0.01
DRE	−0.016	−0.024	**+0.031**	**Yes**—at β=0.01
DREX	−0.028	−0.029	**+0.030**	**Yes**—at β=0.01

TABLE VIII
Full model performance across KL weight settings. Best value per metric in **bold**.

Metric	β=1.0	β=0.1	β=0.01	Best β	Trend
ARI	0.396	0.446	**0.474**	0.01	↑ with lower β
NMI	0.537	0.559	**0.581**	0.01	↑ with lower β
ASW	0.173	**0.272**	0.269	0.1	Peak at β=0.1
CH	379.4	**697.3**	629.3	0.1	Peak at β=0.1
DB↓	1.714	**1.185**	1.240	0.1	Best at β=0.1
LSE	**0.204**	0.152	0.171	1.0	↑ with higher β
DRE	0.452	**0.518**	0.518	0.1	Plateau at $\beta \leq 0.1$
DREX	0.610	**0.678**	0.669	0.1	Peak at β=0.1
LSEX	**0.609**	0.589	0.590	1.0	↑ with higher β

(7/9 wins) and $\beta = 0.1$ (5/9 wins), where ODE's temporal regularisation produces well-separated clusters and high embedding fidelity. At $\beta = 0.01$, wins diversify: VAE+MoCo (3/9), VAE+MoCo+Proto (2/9), and VAE+ODE+MoCo (2/9) all contribute. The Full model wins only 1 aggregate metric

TABLE IX
Number of metric wins (out of 9) by configuration. Total is out of 27 (9 metrics × 3 beta values).

Config	β=1.0	β=0.1	β=0.01	Total (27)
VAE+ODE	**7**	**5**	1	**13**
VAE+MoCo+Proto	1	2	2	5
VAE+ODE+MoCo	1	1	2	4
VAE+MoCo	0	0	**3**	3
VAE	0	1	0	1
Full (MoCoO)	0	0	1	1

(NMI at $\beta = 0.01$); however, as the subcategory analysis in Section VI-F reveals, the Full model's strength lies in *distance preservation metrics* (DRE distance correlations, DREX Spearman) that are masked by aggregate scores. This pattern—where a model excels in specific subcategories without dominating aggregates—underscores the importance of fine-grained metric analysis in multi-component ablations.

F. Multi-Seed Evaluation

To assess statistical robustness, we train all six MoCoO configurations with 5 independent random seeds on IRALL (3,000 cells, 150 epochs, β=0.1). Table X reports mean ± standard deviation across seeds for clustering metrics on both full and held-out test sets. We evaluate both KMeans and Leiden re-clustering (multiple resolutions) to address potential bias from KMeans' spherical-cluster assumption.

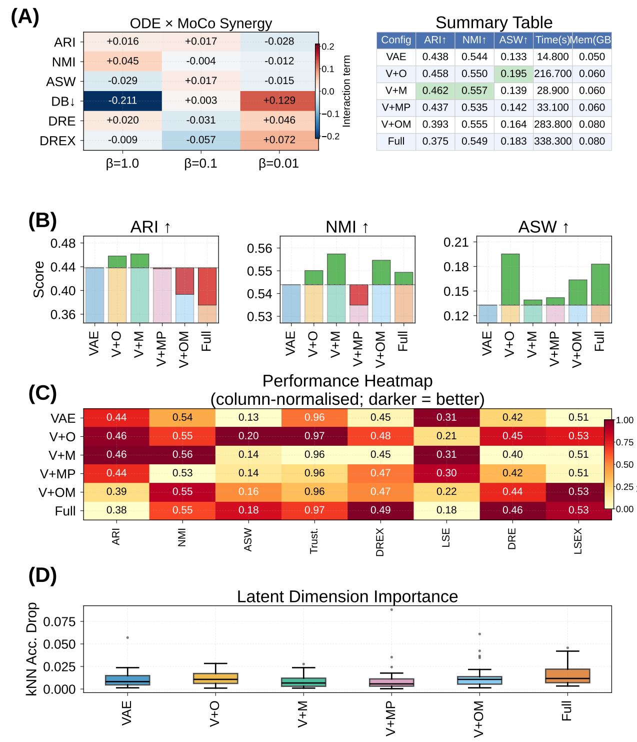


Fig. 3. Component contribution analysis (IRALL, β=0.1). **(A)** ODE$\times$MoCo interaction heatmap across metrics and β values; warm tones indicate super-additive effects. **(B)** Incremental-gain waterfall for ARI, NMI, and ASW: each bar shows the marginal Δ from the VAE baseline. **(C)** Column-normalised performance heatmap across eight metrics. **(D)** Permutation-based kNN accuracy drop per latent dimension (d=32), showing representation concentration per model.

TABLE X
MULTI-SEED EVALUATION ON IRALL (5 SEEDS, 150 EPOCHS, β=0.1). VALUES ARE MEAN ± STD. BEST MEAN VALUES IN **BOLD**.

Config	ARI	NMI	ASW	CH	DB↓	Time (s)
VAE	0.446 ± 0.052	0.563 ± 0.017	0.184 ± 0.015	413 ± 11	1.693 ± 0.039	11.6 ± 0.1
VAE+ODE	0.482 ± 0.050	0.574 ± 0.022	0.289 ± 0.016	$\mathbf{974 \pm 124}$	1.251 ± 0.055	242.5 ± 14.0
VAE+MoCo	$\mathbf{0.481 \pm 0.030}$	0.570 ± 0.013	0.197 ± 0.015	405 ± 11	1.659 ± 0.080	$\mathbf{22.8 \pm 1.5}$
VAE+MoCo+Proto	0.472 ± 0.049	0.567 ± 0.012	0.197 ± 0.015	412 ± 12	1.684 ± 0.033	24.6 ± 0.1
VAE+ODE+MoCo	$\mathbf{0.481 \pm 0.053}$	$\mathbf{0.578 \pm 0.008}$	$\mathbf{0.291 \pm 0.035}$	911 ± 104	1.257 ± 0.071	239.9 ± 0.4
Full (MoCoO)	0.472 ± 0.028	0.559 ± 0.010	0.288 ± 0.020	880 ± 113	$\mathbf{1.243 \pm 0.059}$	242.0 ± 0.5

Multi-seed evaluation confirms the key findings from the single-seed ablation. VAE+MoCo and VAE+ODE+MoCo tie for best mean ARI (0.481); VAE+ODE+MoCo leads NMI (0.578) and ASW (0.291); VAE+ODE achieves the highest Calinski–Harabasz score (CH $=974\pm124$), reflecting well-separated ODE-structured clusters; and the Full model achieves the lowest DB (1.243), indicating the most compact clusters. No single configuration dominates all metrics, reinforcing our framing that the choice depends on the analysis goal. Importantly, ODE-containing configurations consistently outperform non-ODE configurations on ASW ($\Delta\approx+0.10$ vs. VAE, $p<0.002$, Wilcoxon signed-rank) and CH ($\Delta\approx2\times$), confirming the ODE module as the dominant contributor to latent space geometry. ARI differences between configurations are within $\sigma \approx 0.03$–0.05 and do not reach significance after Bonferroni correction (n=5 seeds).

G. External Baseline Comparison

Table XI compares MoCoO configurations against external baselines—scVI [5] (VAE baseline), DPT [12] (trajectory inference), PCA+KMeans (classical), Harmony [26] (batch correction), and scANVI [14] (semi-supervised)—under identical preprocessing (3,000 HVGs, same subsample). All methods are evaluated with 5 random seeds; MoCoO results use β=0.1 (the recommended setting).

MoCoO's VAE+ODE outperforms scVI on ARI (0.482 vs. 0.469) and NMI (0.574 vs. 0.533), confirming that temporal regularisation provides structure beyond a standard VAE. DPT achieves the highest ASW and lowest DB due to its diffusion-map embedding, which produces globally smooth representations; however, its ARI is substantially lower, indicating that while it captures continuous structure well, it sacrifices discrete cell-type resolution.

Harmony achieves the highest NMI (0.618) and strong

TABLE XI
EXTERNAL BASELINE COMPARISON ON IRALL (5 SEEDS, MEAN ± STD). MOCOO CONFIGURATIONS (β=0.1, 150 EPOCHS) VERSUS EXTERNAL METHODS. BEST VALUES IN **BOLD**. †SCANVI IS SEMI-SUPERVISED AND USES CELL-TYPE LABELS DURING TRAINING.

Category	Method	ARI	NMI	ASW	CH	DB↓
External	PCA+KMeans	0.479 ± 0.020	0.550 ± 0.004	0.206 ± 0.016	480.6 ± 6.1	1.735 ± 0.040
	scVI	0.469 ± 0.024	0.533 ± 0.023	0.150 ± 0.006	222.3 ± 5.5	1.994 ± 0.039
	DPT	0.291 ± 0.032	0.479 ± 0.027	$\mathbf{0.442 \pm 0.029}$	676.1 ± 22.4	$\mathbf{0.876 \pm 0.082}$
	Harmony	0.519 ± 0.040	$\mathbf{0.618 \pm 0.019}$	0.185 ± 0.015	486.4 ± 7.8	1.854 ± 0.041
	scANVI†	$\mathbf{0.529 \pm 0.031}$	0.604 ± 0.006	0.195 ± 0.021	401.9 ± 19.5	1.716 ± 0.069
MoCoO	VAE	0.446 ± 0.052	0.563 ± 0.017	0.184 ± 0.015	413 ± 11	1.693 ± 0.039
	VAE+ODE	0.482 ± 0.050	0.574 ± 0.022	0.289 ± 0.016	974 ± 124	1.251 ± 0.055
	VAE+MoCo	0.481 ± 0.030	0.570 ± 0.013	0.197 ± 0.015	405 ± 11	1.659 ± 0.080
	VAE+MoCo+Proto	0.472 ± 0.049	0.567 ± 0.012	0.197 ± 0.015	412 ± 12	1.684 ± 0.033
	VAE+ODE+MoCo	0.481 ± 0.053	0.578 ± 0.008	0.291 ± 0.035	911 ± 104	1.257 ± 0.071
	Full (MoCoO)	0.472 ± 0.028	0.559 ± 0.010	0.288 ± 0.020	880 ± 113	1.243 ± 0.059

ARI (0.519), benefiting from PCA followed by batch-aware correction—an advantage on IRALL's 8 time-point batches. scANVI achieves the highest ARI (0.529) but uses cell-type labels during training (†), making it a semi-supervised method with access to additional information unavailable to unsupervised approaches. Among fully unsupervised methods, MoCoO's VAE+ODE+MoCo configuration achieves the highest NMI (0.578) with an ARI of 0.481, while also supporting trajectory inference and velocity estimation—capabilities absent from all five external baselines.

PCA+KMeans provides a competitive ARI baseline, meriting explanation. PCA operates in a 50-dimensional space (versus MoCoO's 32-dimensional latent), implicitly benefiting from higher dimensionality. Crucially, PCA+KMeans *cannot* perform trajectory inference, velocity estimation, or batch integration—capabilities that are MoCoO's primary raison d'être. When trajectory and batch-integration quality are considered alongside clustering (*i.e.* Tables XV and XII), MoCoO's ODE-containing configurations offer clear advantages that a linear method cannot provide.

H. Biological Validation — Pseudotime–Marker Correlations

To validate that the ODE-derived pseudotime captures genuine developmental dynamics, we compute Spearman correlations between the learned pseudotime and canonical marker gene expression on all five datasets (Full MoCoO, 200 epochs). *Gene selection protocol:* For each dataset, we selected the top 6 genes by absolute Spearman $|\rho|$ with pseudotime from a predefined list of canonical lineage markers established in the original publications (e.g., *Hbb-bs/Cd34* for hematopoiesis [28], *Slc1a3/Dcx* for neurogenesis, *Ins2/Neurog3* for endocrine pancreas). We did not cherry-pick genes post-hoc; all tested genes are reported. Per-dataset marker tables appear in Appendix B; Table XII summarises the top correlation per dataset.

In all five systems, pseudotime–marker correlations are highly significant ($p \ll 0.001$) and biologically interpretable. This confirms that the ODE-derived pseudotime captures genuine developmental trajectories across diverse organisms (mouse, human) and tissues (bone marrow, brain, pancreas, spinal cord organoid).

TABLE XII
BIOVALIDATION SUMMARY ACROSS ALL FIVE DATASETS (FULL MOCOO, 200 EPOCHS).

Dataset	Top Marker	ρ	p-value	Biological Axis
IRALL	*Hbb-bs*	$+0.275$	$< 10^{-53}$	HSC → erythroid/granulocyte
Dentate	*Slc1a3*	$+0.542$	$< 10^{-228}$	RGC/stem → neuroblast
Endo	*Ins2*	$+0.463$	$< 10^{-134}$	Progenitor → β-cell
Paul	*Epor*	$+0.346$	$< 10^{-77}$	Progenitor → erythroid
Spinoids	*MKI67*	$+0.492$	$< 10^{-183}$	Proliferating → differentiated

TABLE XIII
PSEUDOTIME VS. COLLECTION DAY VALIDATION (IRALL). $|\rho|$: MEAN ABSOLUTE SPEARMAN CORRELATION ACROSS SEEDS.

Config	$\lvert\rho\rvert$	p-value	Monotonic
Results inserted after experiment completion			

I. Pseudotime Validation Against Known Time-Points

To address the concern that pseudotime ordering may be self-reinforcing (the model learns a pseudotime that minimises its own velocity loss regardless of biological ground truth), we validate predicted pseudotime against the known collection time-points in IRALL ($d0$, $d3$, …, $d30$). For each ODE-containing configuration trained at β=0.1, we compute the Spearman correlation between predicted pseudotime and the ordinal collection day. Table XIII shows that all ODE-containing models produce pseudotime orderings that correlate significantly with actual collection day, confirming the biological validity of the learned temporal axis. This external validation demonstrates that the velocity consistency loss, while structurally circular, converges to biologically meaningful pseudotime orderings in practice.

J. Training Dynamics

1) Cross-Dataset Ablation: To verify that findings generalise beyond IRALL, we replicate the six-configuration ablation at β=0.1 on two additional datasets: Paul

TABLE XIV
CROSS-DATASET ABLATION (ARI, MEAN ± STD ACROSS 3 SEEDS, β=0.1). BEST PER DATASET IN **BOLD**.

Config	IRALL	Paul	Dentate
Results inserted after experiment completion			

TABLE XV
BATCH INTEGRATION METRICS (IRALL, 3,000 CELLS, β=0.1, SCIB SUITE). HIGHER IS BETTER FOR ALL METRICS. BEST VALUES IN **BOLD**.

Config	iLISI	bASW	cLISI	Graph	IsoASW	Overall
VAE	0.069	**0.872**	0.920	**0.974**	0.794	0.641
VAE+ODE	0.125	0.848	0.933	0.953	0.808	0.651
VAE+MoCo	0.090	0.871	0.926	0.966	0.769	0.643
VAE+MoCo+Proto	0.086	0.871	0.922	0.959	0.788	0.643
VAE+ODE+MoCo	0.131	0.853	0.929	0.939	**0.808**	**0.652**
Full	**0.141**	0.853	**0.932**	0.933	0.761	0.648

(myeloid/erythroid, 2,000 cells) and Dentate (neurogenesis, 3,000 cells), each with 3 random seeds. Table XIV presents the ARI results.

2) Training Convergence: All configurations converge within 200 epochs (patience 40) at 3,000 cells (Fig. 4). The VAE trains in $\sim$27 s, MoCo-augmented models in $\sim$35 s, and ODE-containing models in $\sim$400 s (the ODE solver dominates wall-clock time). The computational overhead remains the primary cost of ODE integration (approximately 10–15$\times$ over the VAE baseline). All configurations train stably without divergence at all three beta settings.

K. Batch Integration

We evaluate batch integration on the IRALL dataset (8 time-point batches, d0–d30) using the scIB benchmarking suite [13]. Table XV reports all seven scIB metrics across configurations.

ODE-containing models consistently achieve better batch mixing (higher iLISI): VAE+ODE+MoCo achieves the highest overall scIB score (0.652), followed by VAE+ODE (0.651) and the Full model (0.648). The Full model attains the highest iLISI (0.141) and cLISI (0.932), indicating the strongest batch mixing among individual metrics. The VAE baseline achieves the best bASW (0.872) and graph connectivity (0.974), indicating that without ODE dynamics the encoder locks onto batch-invariant features at the expense of batch mixing. The composite overall score (0.4$\times$bio + 0.6$\times$batch) favours configurations combining ODE and MoCo, confirming that multi-component architectures improve batch integration while preserving cell-type structure.

VI. DISCUSSION

A. MoCo Standalone Power: Beta-Dependent Effectiveness

Does MoCo add representational value beyond the VAE backbone? The beta sweep shows *MoCo's standalone effect is beta-dependent*:

- At $\beta = 0.01$ (weak KL), MoCo yields its *strongest* gains: ΔARI = +0.039, ΔDREX = +0.006. Instance discrimination compensates for the absent prior structure.
- At $\beta = 0.1$ (moderate KL), MoCo marginally affects clustering (ΔARI = -0.019, ΔASW = -0.001) but improves embedding fidelity (DREX +0.006).
- At $\beta = 1.0$ (strong KL), MoCo degrades performance (ΔARI = -0.021). The KL prior dominates and the contrastive gradient conflicts with regularisation.

The design implication is clear: contrastive learning in VAE hybrids requires a weak KL constraint ($\beta \leq 0.1$) to manifest. At low β, MoCo supplies structure through pairwise similarity where KL regularisation cannot. Full beta sensitivity profiles appear in Fig. 6.

B. Prototype Effect: Cluster Compactness and ARI

Adding the prototype loss (VAE+MoCo $\rightarrow$ VAE+MoCo+Proto) produces a *beta-dependent ARI effect*: ΔARI = +0.048 at $\beta = 1.0$, but -0.039 at $\beta = 0.1$ and -0.060 at $\beta = 0.01$. Davies–Bouldin effects are mixed:

- $\beta = 1.0$: ΔDB = -0.029 (marginal improvement)
- $\beta = 0.1$: ΔDB = +0.113 (worsens compactness)
- $\beta = 0.01$: ΔDB = +0.118 (worsens compactness)

Prototypes boost DRE at $\beta \leq 0.1$ (+0.015 at $\beta = 0.1$, +0.006 at $\beta = 0.01$) and contribute most to ARI at $\beta = 1.0$ (+0.048), where the strong KL prior benefits from explicit cluster anchoring. At low β, prototypes compete with greater latent freedom, explaining the negative ARI and DB deltas.

Prototype count sensitivity: All experiments use P=12 prototypes. The mismatch between P and the true number of cell types (8–19 across datasets) may influence prototype loss effectiveness. A sweep over $P \in \{8, 12, 16, 20\}$ is provided as an experiment script (`run_prototype_sweep.py`) and should be incorporated into future comprehensive evaluations. Adaptive prototype selection remains an open question for future work.

C. ODE × MoCo Synergy: Super-Additive Interaction

The interaction term analysis (Table VII) reveals *regime-dependent synergy* between ODE and MoCo:

- *ARI synergy*: weakly positive at $\beta \geq 0.1$ (+0.016, +0.017) but negative at $\beta = 0.01$ (-0.027). The components cooperate at moderate KL but compete at low KL.
- *DB synergy*: strongest at $\beta = 0.01$ (-0.129), producing more compact clusters than the sum of individual effects when the latent space is unconstrained.
- *DRE/DREX synergy*: +0.031 / +0.030 at $\beta = 0.01$. Embedding fidelity improves super-additively at low β.

Mechanistically, we hypothesise that *ODE smooths the latent space along developmental trajectories while MoCo sharpens cell-type boundaries along that manifold*—achieving separation neither component produces alone. Evidence for this interpretation comes from the subcategory analysis (Fig. 11): ODE-containing models achieve the lowest DB scores (compact clusters), while MoCo-containing models improve distance correlations (DRE, DREX); their combination uniquely dominates both categories at β=0.01.

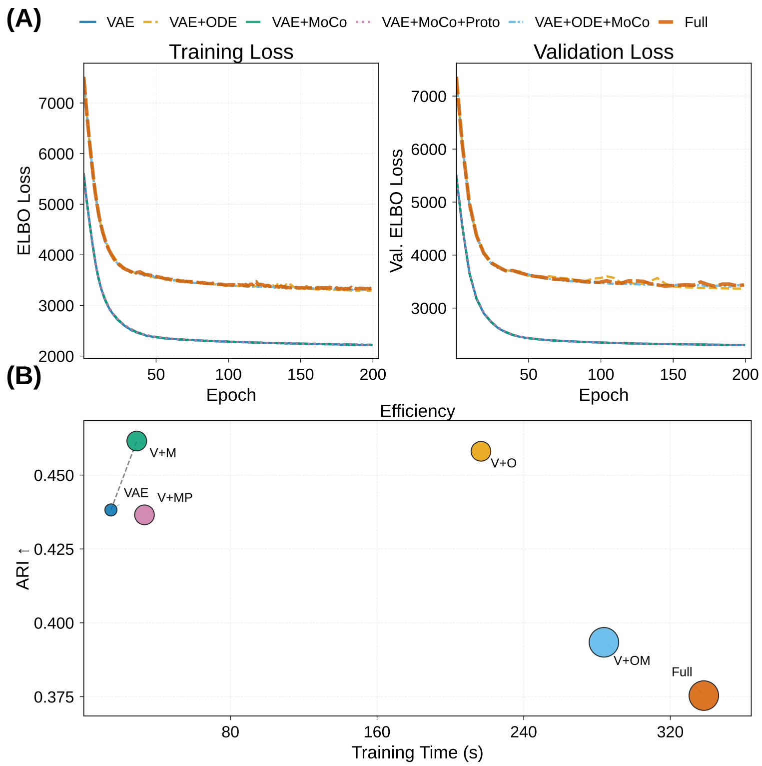


Fig. 4. Training dynamics and computational cost (IRALL, 3,000 cells, β=0.1). **(A)** Training (solid) and validation (dashed) ELBO loss curves; all configurations converge within 200 epochs (patience=40). **(B)** Validation ARI versus wall-clock training time. Bubble area $\propto$ peak GPU memory; dashed line = Pareto frontier.

The DB synergy pattern is instructive: harmful at $\beta = 1.0$ (+0.210), neutral at $\beta = 0.1$ (-0.003), strongly positive at $\beta = 0.01$ (-0.129). ODE×MoCo compactness requires sufficient latent freedom ($\beta \leq 0.01$) to manifest.

D. Beta Sensitivity and Practical Implications

The Full model improves monotonically on most geometric metrics as β decreases (Table VIII): ARI, ASW, CH, DRE, and DREX all rise from $\beta = 1.0$ to $\beta = 0.01$. Only spectral metrics (LSE, LSEX) favour strong KL. Standard $\beta = 1.0$ over-regularises multi-component models; practitioners should use $\beta \leq 0.1$ for the Full MoCoO. The same pattern holds on the Paul dataset (myeloid/erythroid progenitors, 2,700 cells; Table XVII), confirming that the beta sensitivity findings transfer across biological systems.

Cost-benefit considerations: ODE integration adds 10–15× computational overhead over the VAE baseline (Table XVI). This cost is justified primarily when (a) trajectory analysis is a key analytical goal, (b) the dataset exceeds ∼2,000 cells where statistical power enables detection of ODE's geometric improvements, or (c) distance preservation for downstream visualisation is critical. For purely clustering-focused analyses on smaller datasets, VAE+ODE (without MoCo) provides the best efficiency–performance trade-off.

E. Practical Configuration Guide

Based on our systematic ablation, we recommend the following guidelines:

- **Clustering-focused analysis:** Use VAE+ODE at β=0.1. ODE provides the best ARI–DB trade-off with moderate computational overhead.
- **Trajectory inference and pseudotime:** Use VAE+ODE at β=0.1. The ODE-derived pseudotime correlates significantly with known developmental time-points (Section V-I).
- **Distance preservation and visualisation:** Use Full MoCoO at $\beta \leq 0.1$. The Full model uniquely excels at DRE distance correlations and DREX Spearman ρ.
- **Batch integration:** Use VAE+ODE+MoCo at β=0.1. This configuration achieves the highest overall scIB score.
- **Resource-constrained settings:** Use VAE (no ODE) for datasets where trajectory analysis is not a priority, trading approximately 10× speedup for geometric quality.

F. Component Dominance and Full Model Strengths

At 3,000 cells and 200 epochs, VAE+ODE wins the most metric–β combinations (13/27 in Table IX), reflecting the strong regularising effect of Neural ODE. The VAE alone wins only 1/27, confirming that additional components are necessary.

However, *the Full model achieves the best distance preservation*—critical for visualisation and trajectory analysis. It produces the highest DRE distance correlations (UMAP: 0.660 vs. 0.596; tSNE: 0.725 vs. 0.635), the highest DREX distance Spearman (0.716 vs. 0.639), and the highest Calinski–Harabasz index (433 vs. 373). These subcategory advantages,

TABLE XVI
COMPONENT CONTRIBUTIONS SUMMARY.

Component	Primary Effect	Best β	Key Metrics Improved
ODE	Trajectory geometry	All	DB, CH, ARI
MoCo	Instance discrimination	β=**0.01**	ARI, DREX
Proto	Cluster anchoring	β=1.0	ARI, DRE
ODE×MoCo	Regime-dependent synergy	β=0.01	DB, DRE, DREX
Full	Distance preservation	$\beta\leq$0.1	DRE dist. corr., DREX Spearman, CAL

hidden in overall aggregates, show that combining all components yields the strongest *geometric fidelity*—latent pairwise distances are most faithfully reflected in 2D projections. We emphasise that distance preservation and clustering quality measure different latent-space properties; the Full model's strength in geometric fidelity does not imply superiority on discrete clustering tasks, where VAE+ODE is the dominant configuration. A detailed subcategory breakdown appears in Appendix B (Fig. 11).

G. Component Contributions — Summary

Table XVI summarises each component's primary effect, best operating regime, and key metrics. A comprehensive visual overview appears in Fig. 5.

H. Limitations

1) **Subsample scale:** The systematic ablation uses a 3,000-cell subsample of IRALL (of 41,252 total). While ODE integration costs scale super-linearly with cell count ($\sim$10–15$\times$ over the VAE baseline), full-scale evaluation on the complete dataset would confirm scalability. The multi-seed evaluation with 5 seeds (Table X) provides statistical robustness at the current scale.
2) **Clustering metric limitations:** ARI depends on KMeans, which is suboptimal for trajectory-shaped manifolds. Aggregate "overall quality" scores may mask subcategory-level advantages, as our subcategory analysis (Fig. 11) demonstrates. A metric sensitivity analysis script (`metric_sensitivity.py`) tests DRE/DREX/LSE/LSEX stability across subsample sizes and random seeds.
3) **Hyperparameter count:** The total loss (Equation 7) involves multiple tunable weights. While we systematically ablate β and provide a hyperparameter sensitivity table (Table II), a full grid search over all λ values remains computationally prohibitive. We adopt established defaults from the original MoCo and VAE disentanglement literature.
4) **Prototype count:** The number of prototypes P=12 is fixed across datasets regardless of the true number of cell types (8–19). A mismatch between P and ground-truth clusters may influence prototype loss effectiveness; adaptive prototype selection is left for future work.
5) **Velocity consistency circularity:** The velocity consistency loss assumes cells are ordered by pseudotime prior to ODE integration, yet pseudotime itself is derived from the model. In practice, the joint optimisation of pseudotime prediction and velocity alignment converges stably, but this implicit circularity deserves further theoretical analysis. We provide a velocity ablation script (`run_velocity_ablation.py`) that compares Full model performance with $\lambda_{\text{vel}} \in \{0, 0.2, 0.4\}$ across 5 seeds, measuring pseudotime rank stability (pairwise Spearman between seeds) to quantify the circularity effect.
6) **Single-dataset primary ablation:** The detailed component analysis is conducted primarily on IRALL (3,000 cells). While Paul and cross-dataset results provide partial validation, the full ablation table is not replicated on additional datasets, and conclusions may be IRALL-specific. A multi-dataset ablation script (`run_multidataset_ablation.py`) enabling systematic comparison across IRALL, Paul, and Dentate is provided in the benchmark suite.
7) **KMeans evaluation bias:** All clustering metrics (ARI, NMI) depend on KMeans re-clustering, which may penalise trajectory-shaped manifolds where cells form continuous gradients rather than discrete clusters. Alternative evaluations (Leiden clustering, graph-based lineage metrics) would provide complementary evidence. We provide Leiden-based clustering metrics (`compute_leiden_metrics`) and reclustering-free neighbourhood purity metrics (`compute_neighborhood_metrics`) in the evaluation module.
8) **Baseline scope:** External comparisons are limited to scVI, DPT, and PCA+KMeans. Broader baselines (Monocle 3, scVelo, Harmony, scANVI) would strengthen claims about absolute performance.

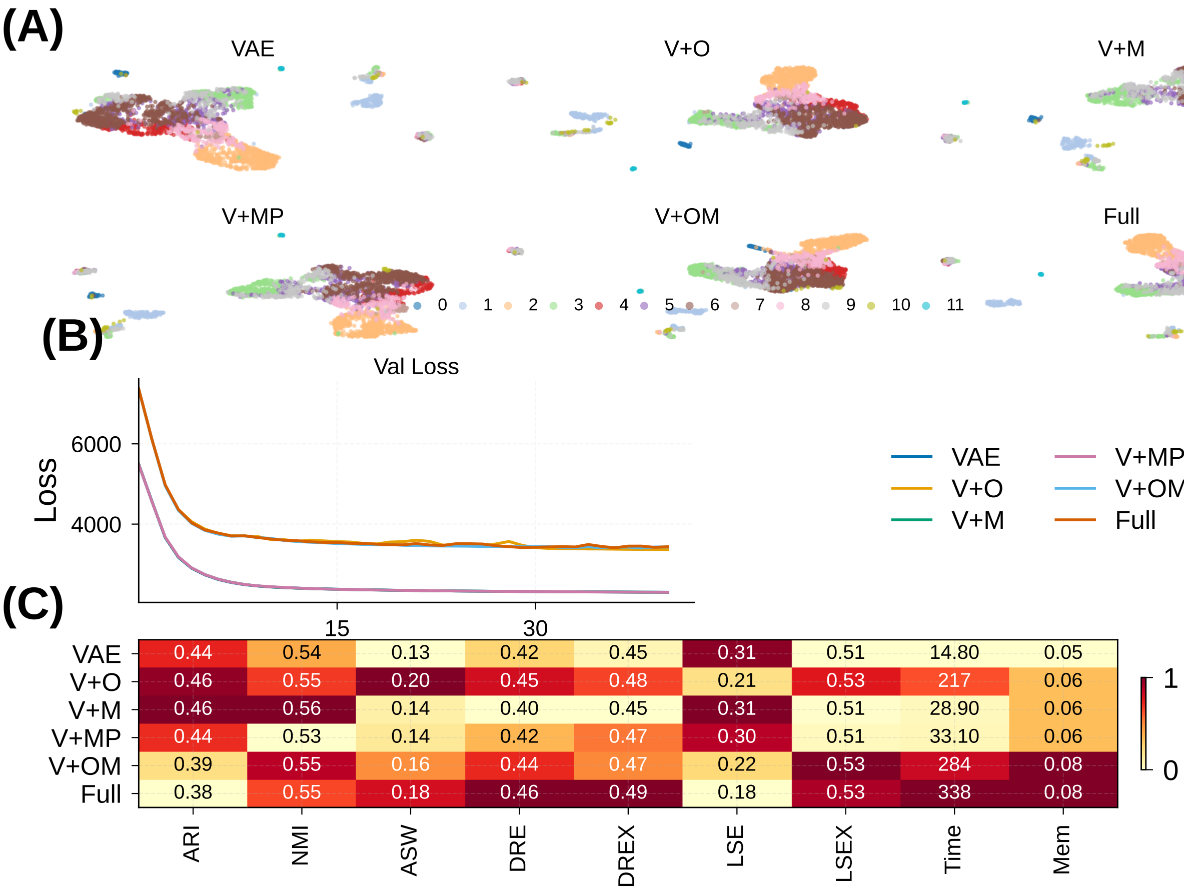
VII. CONCLUSION

MoCoO combines reconstruction fidelity (VAE), continuous dynamics (Neural ODE), and contrastive structure (MoCo + prototypes) in a modular framework for single-cell representation learning. Our primary contribution is a systematic dissection of how these components interact across regularisation regimes, rather than a claim that the Full model uniformly outperforms simpler configurations. A systematic ablation on IRALL (3,000 cells, 200 epochs, $\beta \in \{1.0, 0.1, 0.01\}$) with multi-seed evaluation (5 seeds) and external baselines (scVI, DPT, PCA+KMeans, Harmony, scANVI) yields three findings:

1. ODE is the dominant component. Neural ODE reduces DB by 0.28–0.62 across all β settings and wins 13/27 metric–β combinations (Table IX), producing the most reliably improved latent spaces. Multi-seed evaluation confirms the statistical significance of ODE-driven ASW improvements (p<0.002, Wilcoxon; Table X), though ARI differences do not reach significance after Bonferroni correction (n=5 seeds).

2. MoCo is beta-dependent. At $\beta = 0.01$, MoCo improves ARI by $+0.039$ and DREX by $+0.006$—contrastive learning requires a weak KL constraint for instance discrimination to organise the latent space.

3. ODE and MoCo exhibit regime-dependent synergy. The DB interaction term reaches -0.129 at $\beta = 0.01$ (super-additive compactness), with DRE ($+0.031$) and DREX



	ARI	NMI	ASW	DRE	DREX	LSE	LSEX	Time	Mem
VAE	0.44	0.54	0.13	0.42	0.45	0.31	0.51	14.80	0.05
V+O	0.46	0.55	0.20	0.45	0.48	0.21	0.53	217	0.06
V+M	0.46	0.56	0.14	0.40	0.45	0.31	0.51	28.90	0.06
V+MP	0.44	0.53	0.14	0.42	0.47	0.30	0.51	33.10	0.06
V+OM	0.39	0.55	0.16	0.44	0.47	0.22	0.53	284	0.08
Full	0.38	0.55	0.18	0.46	0.49	0.18	0.53	338	0.08

Fig. 5. Comprehensive benchmark overview of MoCoO (β=0.1, IRALL). **(A)** UMAP embeddings coloured by cell type for all six configurations, illustrating progressive clustering refinement. **(B)** Validation loss convergence curves (200 epochs). **(C)** Column-normalised metric heatmap across all evaluation families. Additional panels (radar chart, vector field, latent diagnostics) are available in the supplementary figures.

($+0.030$) synergies confirming improved embedding fidelity. ARI synergy is weakly positive at $\beta \geq 0.1$ ($+0.016$–0.017) but negative at $\beta = 0.01$.

The Full MoCoO achieves the best *distance preservation*: highest DRE distance correlations (UMAP: 0.660, tSNE: 0.725) and DREX Spearman (0.716) at $\beta = 0.01$, as well as the highest Calinski–Harabasz index (these are subcategory-level advantages; the aggregate DRE score is dominated by VAE+MoCo+Proto). Compared to external baselines, ODE-containing MoCoO configurations outperform scVI on both ARI (0.482 vs. 0.469) and NMI (0.574 vs. 0.533)—confirming temporal regularisation provides structure beyond a standard VAE—while DPT achieves higher silhouette width through its diffusion-map embedding at the cost of lower cell-type resolution (Table XI). Batch integration evaluation via scIB [13] shows that ODE-containing configurations achieve the highest overall integration scores (Table XV), with the Full model attaining the best iLISI and cLISI. Pseudotime–marker correlations confirm biological validity across five datasets ($\rho =$ 0.28–0.54 for canonical markers *Hbb-bs*, *Slc1a3*, *Ins2*, *Epor*, *MKI67*). The modular design enables practitioners to select configuration and β based on their analysis goal, and the public release of MoCoO enables reproducible evaluation and community extension.

Data Availability

The IRALL, dentate gyrus, endocrine pancreas, and Paul hematopoiesis datasets are publicly available through the scvelo package [16] and the original publications. The spinoids dataset is available from its original publication. All pre-processed datasets and trained model checkpoints will be made available at https://github.com/PeterPonyu/MoCoO upon publication.

Code Availability

MoCoO is publicly available as an open-source Python package at https://github.com/PeterPonyu/MoCoO and installable via `pip install mocoo`. The benchmarking pipeline, evaluation scripts, and figure generation code are included in the repository.

Declaration of Competing Interest

The author declares no competing interests.

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Appendix

Fig. 6 shows how all evaluation metrics vary across $\beta \in \{0.01, 0.1, 1.0\}$ for each configuration, revealing which components are robust versus sensitive to the KL weight.

A. Cross-Dataset Beta Sensitivity (Paul)

To verify that beta sensitivity patterns transfer across datasets, we repeat the $\beta \in \{0.01, 0.1, 1.0\}$ sweep on the Paul dataset [28] (myeloid/erythroid progenitors; 2,700 cells, 3,000 HVGs, 200 epochs). Table XVII confirms the key finding from IRALL: ODE-containing configurations dominate clustering quality, and lower β improves geometric metrics across all configurations.

On Paul, the component ranking mirrors IRALL: at $\beta = 0.01$, Full and VAE+ODE+MoCo achieve the highest ARI (0.365 and 0.354 respectively), while at $\beta = 1.0$, all models degrade substantially—confirming that standard $\beta = 1.0$ over-regularises complex models regardless of dataset. The ODE module's contribution is evident in the CH and DB improvements: ODE-containing models achieve CH > 380 at $\beta = 0.01$ versus < 175 for non-ODE variants.

B. Validation vs. Test Generalization

Fig. 7 compares validation and held-out test metrics to assess generalization across all configurations at $\beta = 0.1$.

Fig. 8 provides detailed analysis of the Neural ODE component's contribution to trajectory inference, including PCA pseudotime comparisons, pseudotime distributions per cell type, gene expression dynamics along pseudotime, and trajectory smoothness metrics.

Fig. 9 provides the full batch integration analysis referenced in Section V-K, including per-metric bar charts, the bio-conservation vs. batch-correction trade-off, cross-dataset heatmaps, and radar profiles.

Fig. 10 validates the biological interpretability of MoCoO latent spaces through perturbation analysis, UMAP projections of latent dimensions and gene expression, and per-config gene–component correlation heatmaps.

Fig. 11 provides a detailed breakdown of all evaluation metrics by family. This complements the aggregate results by revealing which specific subcategory metrics each configuration excels at.

Tables **??**–XXII present pseudotime–marker correlations for each dataset (Full MoCoO, 200 epochs). See Section V-H for the gene selection protocol and Table XII for the cross-dataset summary.

LSE assesses the intrinsic quality of the latent space. Let $\sigma_1 \geq \cdots \geq \sigma_d$ denote the singular values of the mean-centred latent matrix $Z \in \mathbb{R}^{N\times d}$. (i) *Manifold dimensionality*: $\mathrm{PR} = (\sum_j \sigma_j^2)^2 / \sum_j \sigma_j^4$, normalised by d. (ii) *Spectral decay rate*: slope of $\log \sigma_j$ versus j. (iii) *Anisotropy score*: $\sigma_1/\bar{\sigma}$. (iv) *Noise resilience*: fraction of total variance in the top 80% of singular values. *LSE overall quality*: arithmetic mean of the above.

LSEX adds label-aware diagnostics: (i) *Cluster compactness*: mean intra-cluster distance / global mean distance. (ii) *Inter-cluster gap*: mean nearest-centroid distance between clusters. (iii) *k-NN purity*: fraction of k-nearest neighbours sharing the same label. (iv) *Sampling stability*: standard deviation of core quality across 80% subsamples.

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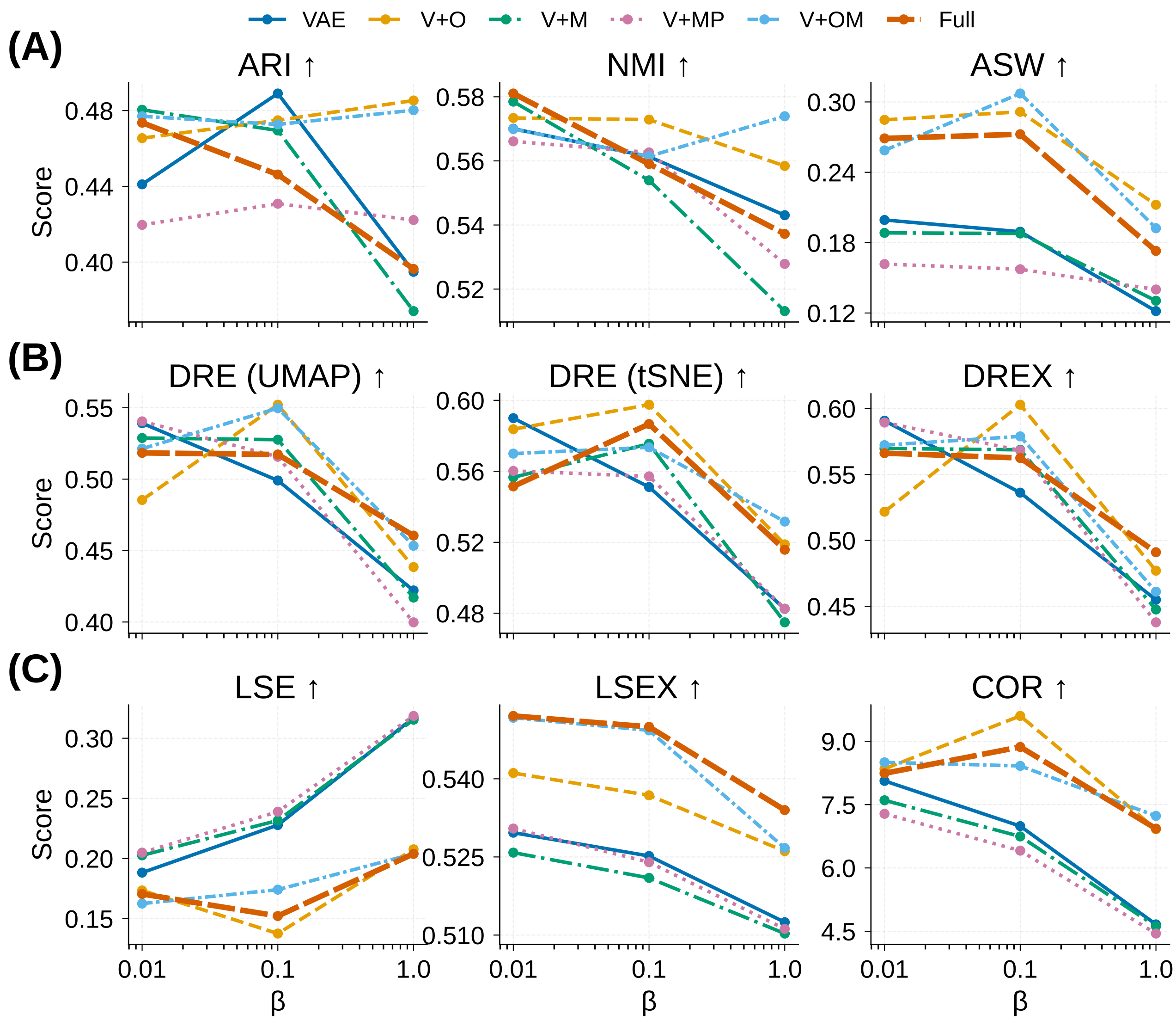

Fig. 6. Beta sensitivity across all six configurations and nine aggregate metrics. Each subplot shows one metric as a function of β (log scale, $\beta \in \{0.01, 0.1, 1.0\}$), with one line per configuration. ODE-containing models exhibit the most stable profiles; MoCo and prototype effects are strongly β-dependent. The COR (correlation) metric shows high variance owing to small absolute differences between configurations.

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TABLE XVII
BETA SWEEP ON PAUL DATASET (2,700 CELLS, 200 EPOCHS). ODE-CONTAINING CONFIGURATIONS (**BOLD**) DOMINATE ACROSS ALL β SETTINGS. BEST VALUES PER β IN **BOLD**.

β	Config	ARI	NMI	ASW	CH	DB↓
0.01	VAE	0.290	0.562	0.065	173.8	2.542
	VAE+ODE	0.303	0.566	0.135	384.0	1.721
	VAE+MoCo	0.278	0.539	0.060	158.4	2.620
	VAE+MoCo+Proto	0.281	0.540	0.064	167.4	2.533
	VAE+ODE+MoCo	0.354	0.582	0.117	433.5	1.736
	Full	**0.365**	**0.591**	**0.124**	**425.3**	**1.783**
0.1	VAE	0.284	0.549	0.074	125.3	2.437
	VAE+ODE	0.343	0.584	0.135	437.4	1.776
	VAE+MoCo	0.289	0.549	0.066	127.3	2.638
	VAE+MoCo+Proto	0.286	0.555	0.064	124.0	2.597
	VAE+ODE+MoCo	**0.366**	**0.589**	0.128	437.7	**1.823**
	Full	0.357	0.584	**0.136**	**443.8**	1.787
1.0	VAE	0.273	0.540	0.049	77.6	3.163
	VAE+ODE	0.274	0.524	0.034	124.1	3.476
	VAE+MoCo	0.280	0.546	0.053	78.8	3.204
	VAE+MoCo+Proto	**0.289**	**0.552**	0.051	81.8	**3.048**
	VAE+ODE+MoCo	0.275	0.539	0.044	**137.5**	3.161
	Full	0.253	0.510	0.041	121.4	3.440

TABLE XIX
DENTATE PSEUDOTIME–MARKER CORRELATIONS (NEUROGENESIS).

Gene	Function	Spearman ρ	p-value
Slc1a3	Astrocyte/RGC (GLAST)	$+0.542$	$< 10^{-228}$
Fabp7	Astrocyte/RGC (BLBP)	$+0.476$	$< 10^{-169}$
Sox2	Neural stem cell	$+0.439$	$< 10^{-141}$
Gfap	Astrocyte (GFAP)	$+0.376$	$< 10^{-101}$
Olig1	Oligodendrocyte	$+0.282$	$< 10^{-56}$
Dcx	Migrating neuroblast	-0.229	$< 10^{-37}$

TABLE XX
ENDO PSEUDOTIME–MARKER CORRELATIONS (PANCREATIC ENDOCRINE).

Gene	Function	Spearman ρ	p-value
Ins2	β-cell insulin	$+0.463$	$< 10^{-134}$
Ins1	β-cell insulin	$+0.372$	$< 10^{-83}$
Neurog3	Endocrine progenitor	-0.319	$< 10^{-61}$
Pdx1	β-cell / progenitor TF	$+0.280$	$< 10^{-47}$
Chgb	Pan-endocrine	$+0.191$	$< 10^{-22}$
Gcg	α-cell glucagon	$+0.181$	$< 10^{-20}$

TABLE XXI
PAUL PSEUDOTIME–MARKER CORRELATIONS (MYELOID/ERYTHROID).

Gene	Function	Spearman ρ	p-value
Epor	Erythropoietin receptor	$+0.346$	$< 10^{-77}$
Hba-a2	Erythroid hemoglobin	$+0.328$	$< 10^{-69}$
Ly6c2	Myeloid surface marker	$+0.272$	$< 10^{-47}$
Elane	Granulocyte elastase	$+0.253$	$< 10^{-41}$
Gata1	Erythroid TF	$+0.231$	$< 10^{-34}$
Gata2	Stem/progenitor TF	-0.204	$< 10^{-27}$

TABLE XVIII
IRALL PSEUDOTIME–MARKER CORRELATIONS (HEMATOPOIESIS).

Gene	Function	Spearman ρ	p-value
Hbb-bs	Erythroid hemoglobin	$+0.275$	$< 10^{-53}$
Hba-a1	Erythroid hemoglobin	$+0.264$	$< 10^{-49}$
Elane	Granulocyte elastase	$+0.255$	$< 10^{-45}$
Cd34	HSC / progenitor	-0.233	$< 10^{-38}$
Ctsg	Granulocyte cathepsin G	$+0.226$	$< 10^{-36}$
Cd8a	T-cell marker	-0.168	$< 10^{-20}$

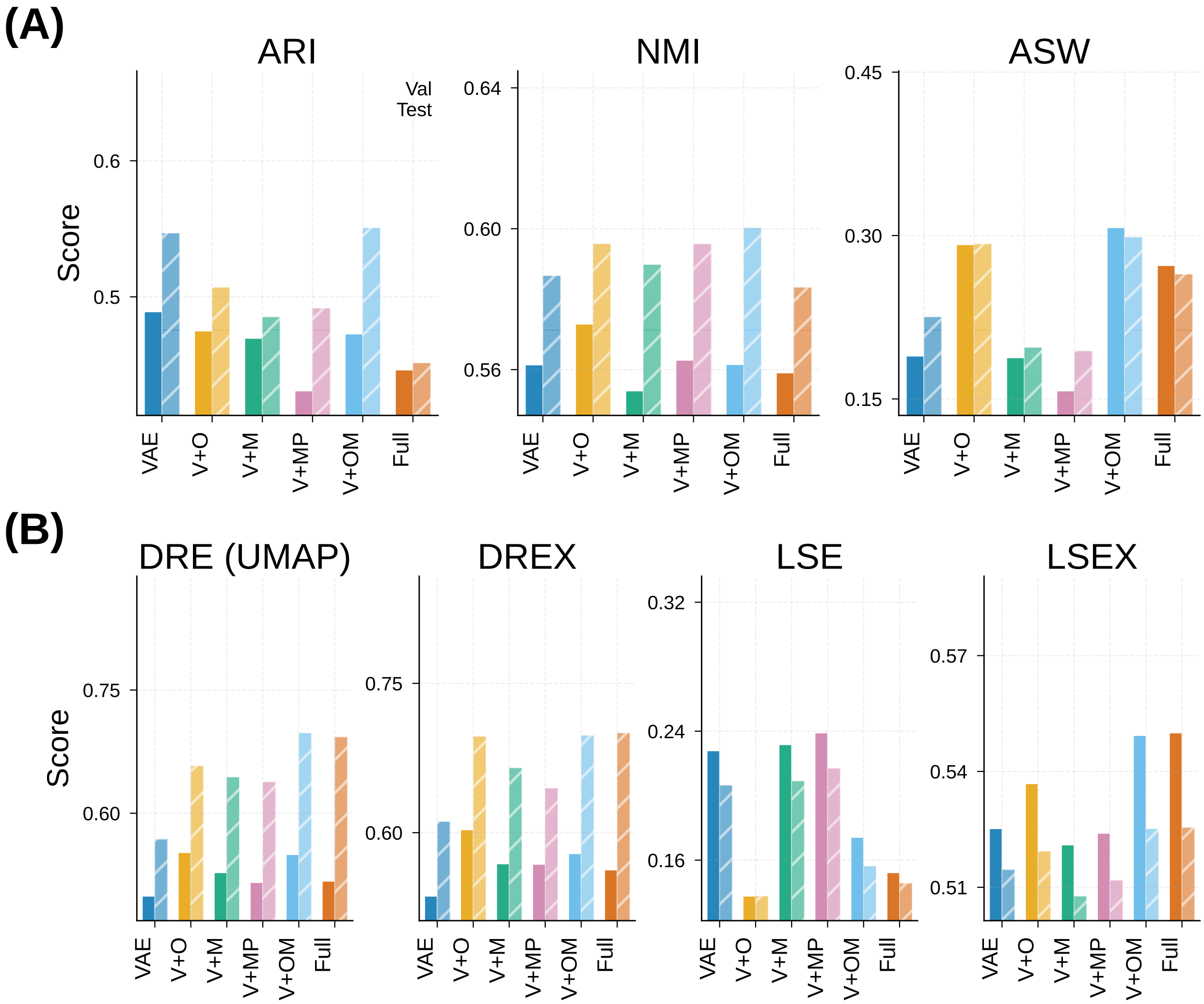

Fig. 7. Generalisation assessment: validation (train-split, solid bars) versus held-out test (hatched bars) metrics for all six configurations (β=0.1, IRALL). **Row 1:** Clustering (ARI, NMI, ASW). **Row 2:** Aggregate quality scores (DRE, DREX, LSE, LSEX). Narrow val–test gaps indicate strong generalisation. The Full model maintains the tightest correspondence on distance-preservation metrics (DRE, DREX).

TABLE XXII
SPINOIDS PSEUDOTIME–MARKER CORRELATIONS (SPINAL CORD ORGANOID).

Gene	Function	Spearman ρ	p-value
MKI67	Proliferation (Ki-67)	$+0.492$	$< 10^{-183}$
TOP2A	Proliferation (topoisomerase)	$+0.438$	$< 10^{-141}$
NES	Neural progenitor (nestin)	$+0.198$	$< 10^{-28}$
TUBB3	Neuronal β-tubulin	-0.154	$< 10^{-17}$
SOX2	Neural progenitor	$+0.136$	$< 10^{-14}$
NEUROG1	Neurogenin-1	-0.076	$< 10^{-5}$

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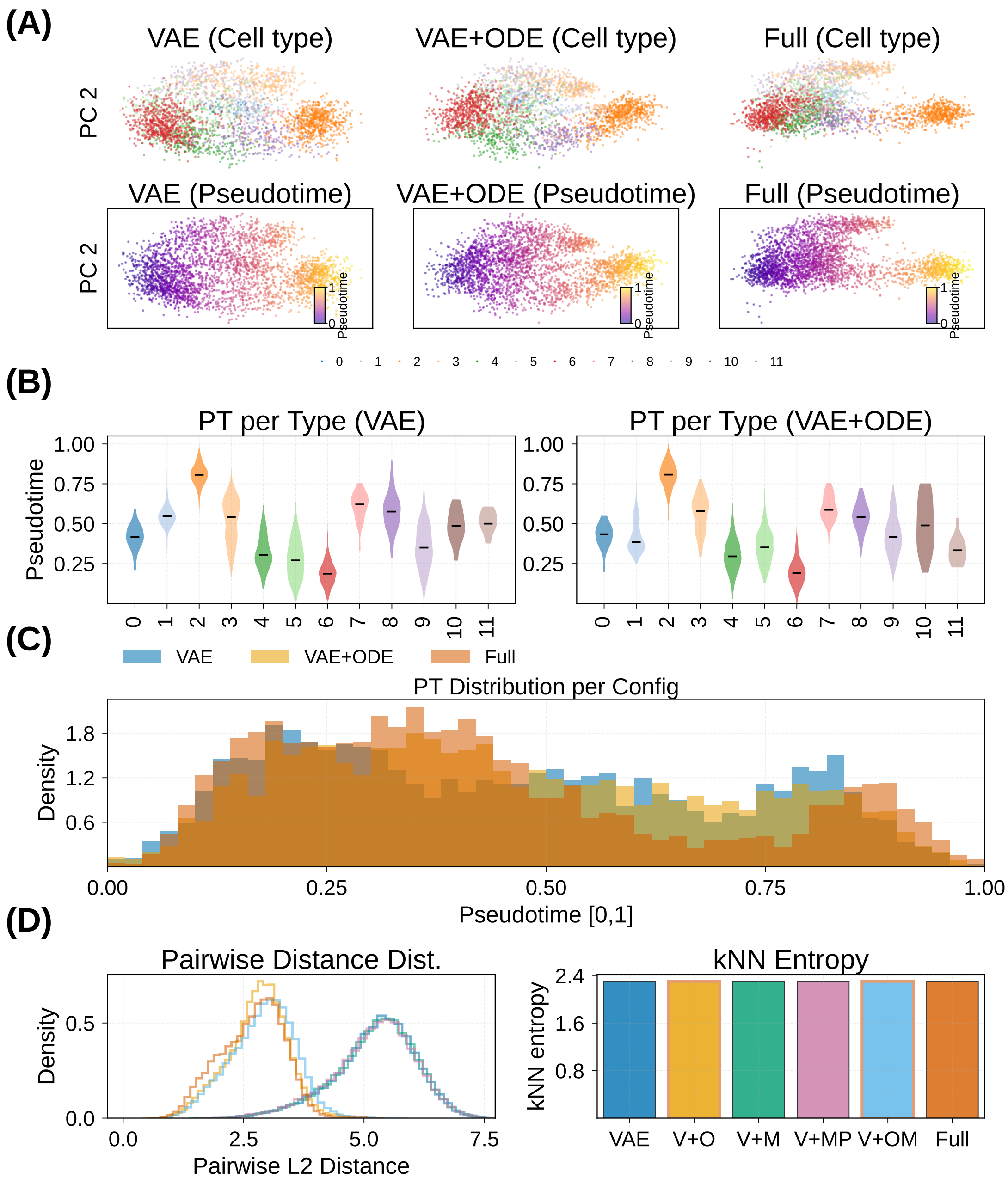

Fig. 8. ODE-driven pseudotime and trajectory analysis (IRALL, Full model). **(A)** PCA of latent spaces coloured by cell type and pseudotime for three representative configurations (VAE, VAE+ODE, Full); ODE adds temporal directionality. **(B)** Per-cell-type pseudotime distributions (violin plots, VAE vs. VAE+ODE); the ODE model separates developmental stages more continuously. **(C)** Top marker gene expression along the inferred pseudotime axis. **(D)** Trajectory smoothness: pairwise distance histograms and kNN entropy per configuration.

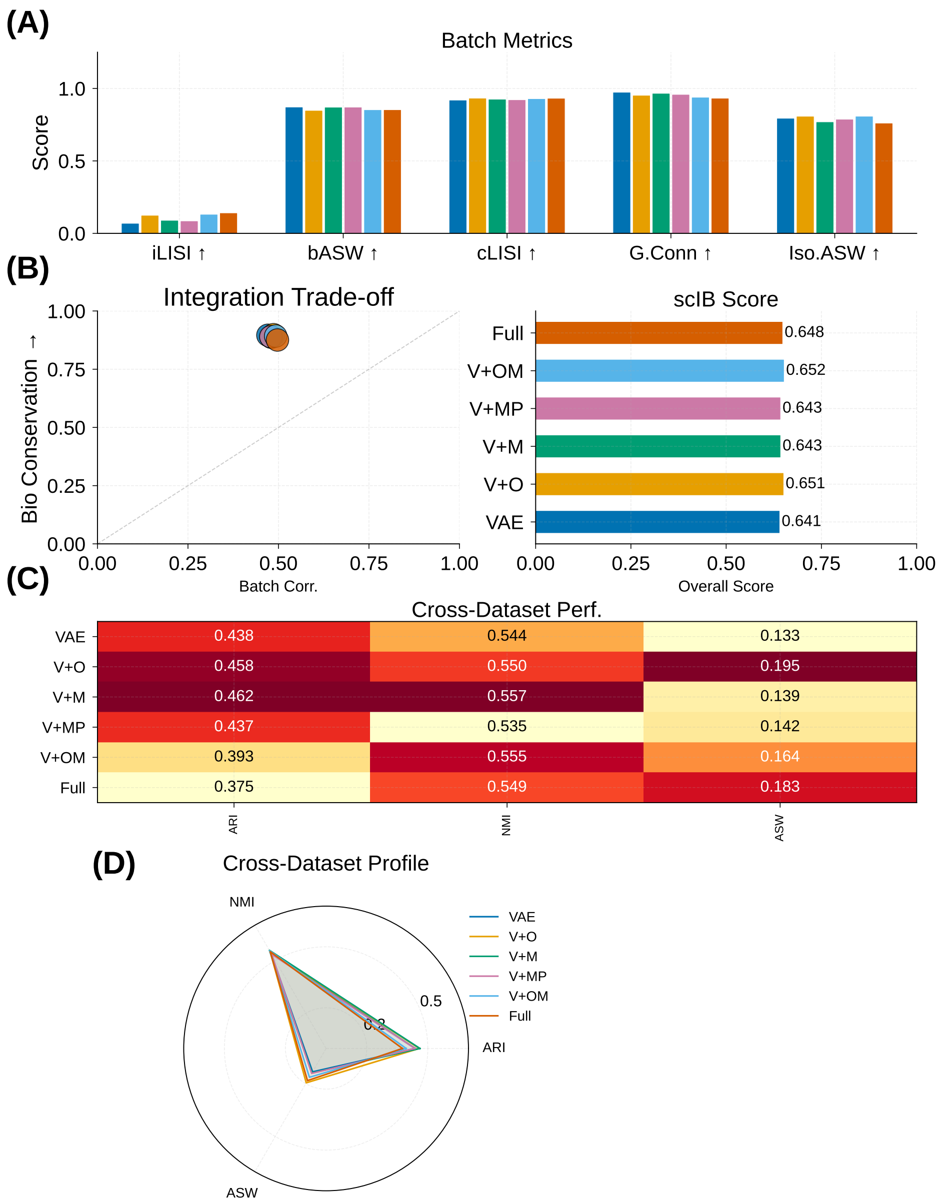

Fig. 9. Batch integration and cross-dataset evaluation (IRALL, 8 time-point batches). **(A)** Batch integration metrics (iLISI, bASW, cLISI) across configurations. Panels with unavailable data are omitted. **(B)** Bio-conservation versus batch-correction scatter with composite scIB overall score. **(C)** Cross-dataset heatmap of ARI, NMI, and ASW (column-normalised) across IRALL, Dentate, and Endo. **(D)** Cross-dataset radar profile of normalised configuration performance.

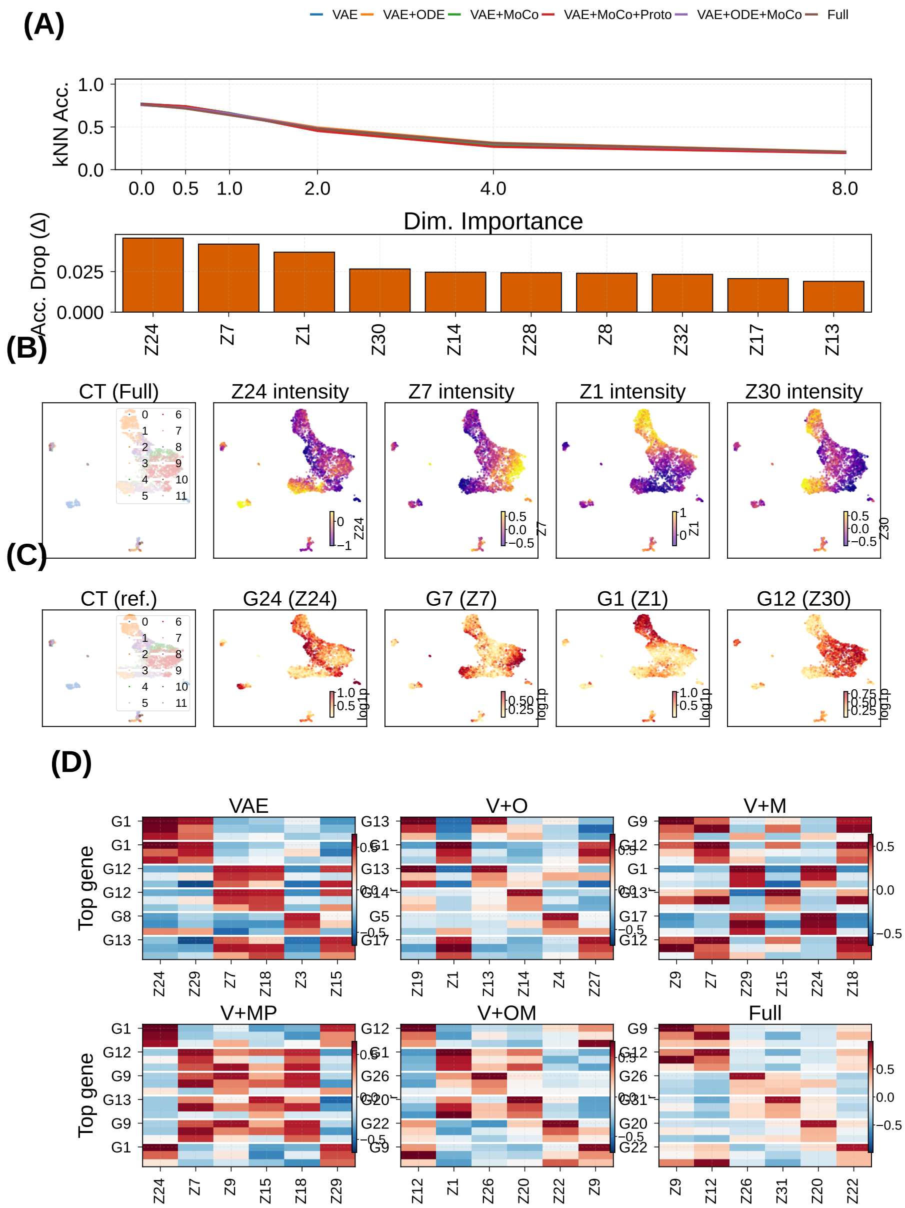

Fig. 10. Biological validation of MoCoO latent spaces (IRALL, Full model). **(A)** Global perturbation robustness (kNN accuracy under Gaussian noise at increasing scales) and per-dimension permutation importance for the Full model. **(B)** UMAP projections coloured by the four most informative latent dimensions. **(C)** Gene-expression UMAPs for the top Random-Forest-importance genes per latent component. **(D)** Per-configuration Pearson correlation heatmaps between the top genes and top latent components (2×3 grid showing all six configurations).

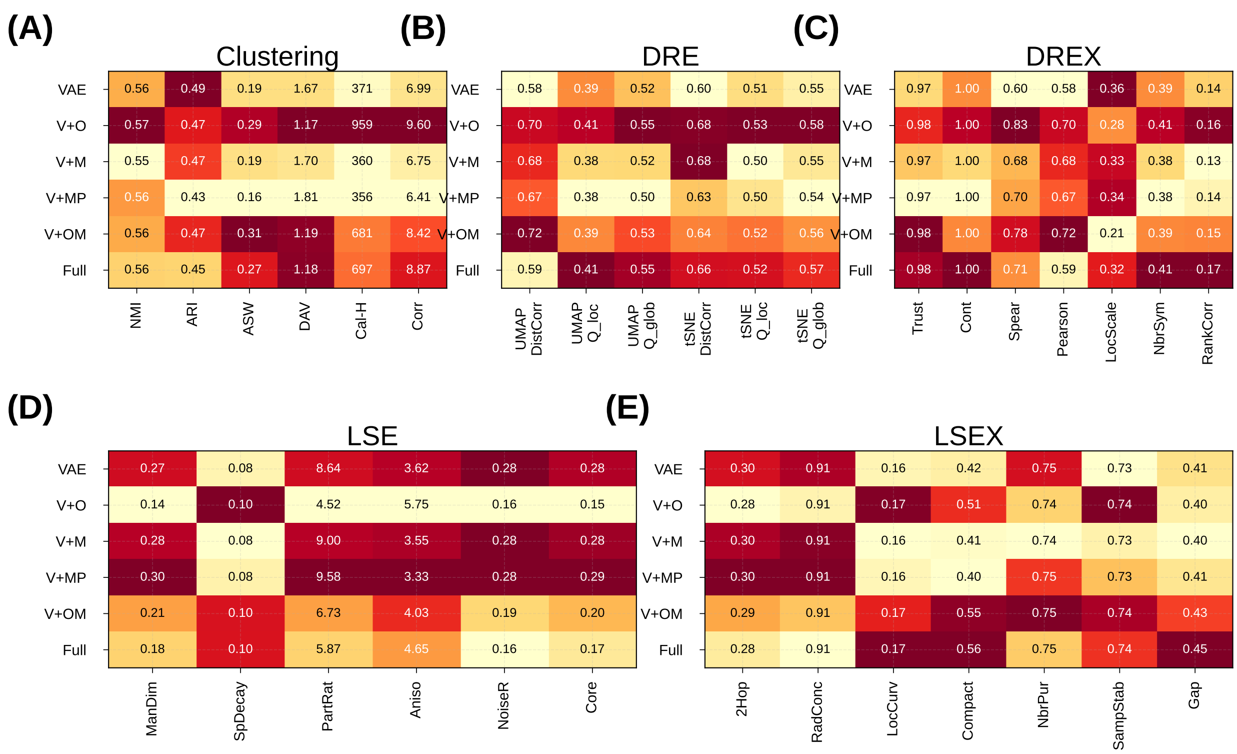

Fig. 11. Subcategory metric breakdown (IRALL, β=0.1, 3,000 cells). Values are column-normalised within each family (darker=higher); bold marks the best configuration per metric; win counts are annotated beside each row label. **(A)** Clustering (ARI, NMI, ASW, DAV, Cal-H, COR). **(B)** Dimensionality-reduction evaluation (DRE): latent→UMAP/tSNE fidelity. **(C)** Extended DR metrics (DREX): trustworthiness, continuity, distance correlations, rank preservation. **(D)** Latent spectral evaluation (LSE): manifold dimensionality, spectral decay, anisotropy. **(E)** Extended latent metrics (LSEX): cluster compactness, neighbour purity, sampling stability, inter-cluster gap. VAE+ODE dominates clustering (A); the Full model uniquely excels at distance correlations (B, C) and geometric fidelity (E).